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TITLE:

ANTI-GRAVITY #3

Force Field Shows Propulsion Promise

SAN DIEGO—A promising research approach to force field propulsion is reported by the Electronics Div., Ryan Aeronautical Co., San Diego, Calif.

The system is being studied for possible application in a series of manned watch satellites for detection and intercept of hostile ICBM's.

Martin N. Kaplan, a senior research engineer at Ryan Electronics, does not think present-day chemically fueled rockets have the duration required for an orbital mission. More exotic propulsion methods—nuclear, ion or plasma-type systems—can be ruled out because of weight or very low initial velocities, he says.

The need, according to Kaplan, is for a new approach to propulsion which would furnish very high speeds along with high initial velocities and long duration.

Ryan research in force fields is running along two paths that "look promising." One is termed "unified field theory," the other a "quantum mechanical theory of free space." The latter approach, derived from a quan-

tized model, has led to successful experiments, Kaplan says. Additional experiments are being set up.

A more comprehensive research program is planned which, if successful, will produce either "restricted" or "general" results. Either would be a revolution in propulsion.

Restricted results might be dubbed antigravitation. This would permit an acceleration force which could be directed only toward, or away from, a second body. A general result would permit a reaction against free space.

The general result is anticipated. But this would still require one major breakthrough—development of a means of converting potential energy (perhaps nuclear) to kinetic energy in the form of a rectilinear momentum without expelling mass. Such a breakthrough seems possible, Kaplan believes, when there is a better understanding of the mechanism by which matter is converted to energy.

• **100-g acceleration**—In practice, the thrust produced by engines of this type would be directly proportional to

their energy conversion rates. Neglecting energy losses, the vehicle's rate of change of momentum would be limited only by its peak energy conversion rate.

Kaplan believes these conversion rates could be safely employed to achieve space accelerations of 100 g or more. This is not so fantastic when compared to recent ultracentrifuge tests which have produced reactions against space of nearly 500,000 g. Once requirements for rectilinear reactions against space are discovered, 100-to-1000-g propulsion should be attainable.

A complete cycle of acceleration and deceleration at 100 g would take a vehicle from the earth's surface to a 1000-mile altitude in 80.8 sec., disregarding atmospheric drag. Although this extreme acceleration rate might be no problem for the vehicle structure, it would hardly create a friendly environment for its occupants or their instruments.

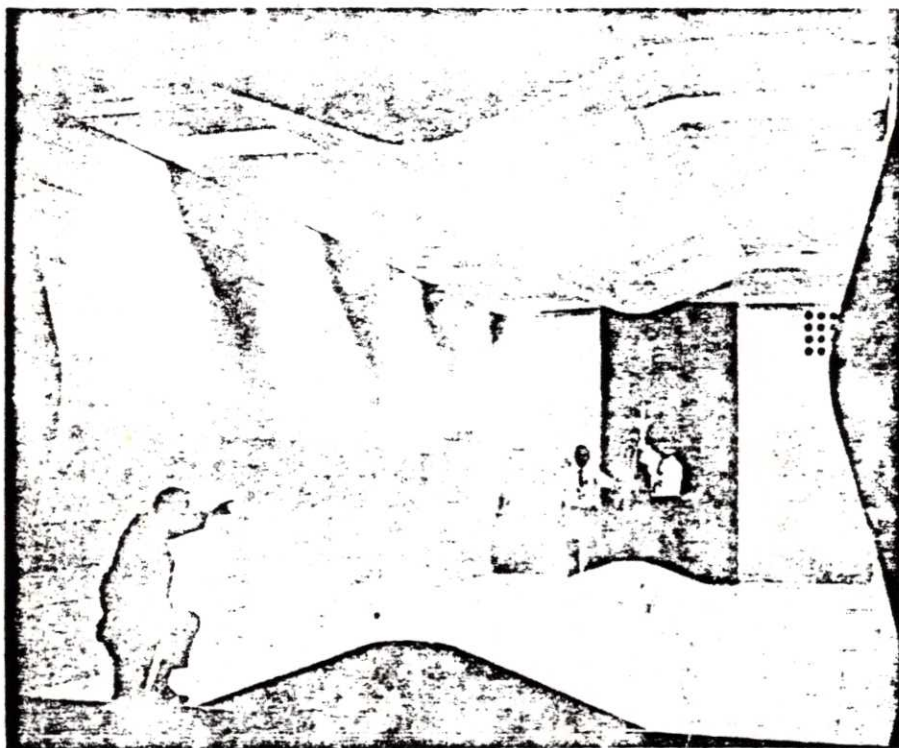
The same principle used for vehicle propulsion could be used to give the vehicle's occupants a g-controlled environment of a less disastrous character.

• **Putting it to work**—So where's the practical use if it all works? Kaplan proposes a system of six "stacked" counterorbiting fleets of 24 manned ICBM detection-pursuit vehicles per orbit and counterorbit. Vehicles in each orbit (12 in each direction) would be spaced about 2075 miles apart. Each fleet would patrol an area at least 2000 miles wide, extending completely around the world. Thus, 144 patrol vehicles would have to be in space at all times to give continuous early warning coverage of the entire earth's surface.

If, for some reason, orbiting vehicles could not be used, the vehicles might be placed in a hovering position, deployed to form a space blockade about a potentially hostile power.

The early warning system could double as repeater stations in closed communication links, separated by slightly over 2000 miles. Tandem vehicles of each orbit, or the vehicles of a hovering blockade, would be within line-of-sight of each other. Thus, they could relay information originating in any one station or from an earth station in contact with any one space point.

Method in Madness



BULGES AND BUCKLES with a purpose were built into this room at Republic Aviation's new Farmingdale, N.Y., Research and Development Center. Gibson-Girl build of walls makes almost completely echo-free environment for testing antennas.

"ELECTRO-GRAVITICS": WHAT IT IS—OR MIGHT BE*

BY A. V. CLEAVER,† F.R.Ae.S., FELLOW

During the past two years, there can be few people in any way interested in either aeronautics or astronautics who have not encountered the unfamiliar term "electro-gravitics," and reacted to it with perplexity, amusement, scepticism, or perhaps a mixture of all three of these attitudes.

Those within the aircraft and guided weapons industries have probably been asked, as I certainly have, "What is the U.K. doing in this field?" by visiting Americans, who apparently find it improbable that Britain, with its tradition of scientific discovery and invention, is inactive in such a new subject. The reply, "Nothing, as far as I know—at least in the sense you mean," is usually received with some surprise, while the alternative, "What is 'electro-gravitics'?" which I have also heard given, is apt to provoke the further inevitable question, "Are you kidding?"! It is not too difficult to understand this, for in the past a familiar pattern has been for new scientific advances to be pioneered over here, or on the Continent, and then exploited by our friends across the Atlantic; the basic science has been mainly European, even when the technology has become predominantly American, at least in its later stages. Perhaps this sequence will not be so often the case in future, but any departure from it still occasions some comment.

It has by no means been necessary to circulate within any privileged circles to have met this strange new word "electro-gravitics." In the United States, newspapers and magazines, and nearly all the American aviation periodicals, have printed references to current work. Some of these accounts have been repeated in British and European papers, again more especially in the aeronautical publications, and sometimes with embellishments. In a few instances, the subject has been dealt with in a style of pseudo-scientific gobbledy-gook which has been worse than meaningless. The only comment called for on these latter is that they obviously emanate from people who realize that here is a good technical "story," possibly even a very important one, but who do not have the facts to write it up properly and who furthermore, to put it bluntly, have not the faintest idea what it is really all about.

American Research on "Electro-Gravitics"

What are the facts, insofar as they are publicly known, or (as at this date) knowable? Well, they seem to amount to this: The Americans have decided to look into the old science-fictional dream of gravity control, or "anti-gravity," to investigate, both theoretically and (if possible) practically the fundamental nature of gravitational fields and their relationship to electromagnetic and other phenomena—and someone (unknown to the present writer) has apparently decided to call all this study by the high-sounding name of "electro-gravitics."

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Unknown, too—or at least unannounced—is the name of agency or individual who decided to encourage, stimulate, or sponsor this effort, also in just what way it is being done. However, that the effort is in progress there can be little doubt, and, of course, it is entirely to be welcomed. One's only legitimate regret might be that, colloquially speaking, the whole business seems to have gone off "half-cock," in that some of the reports already referred to, with their facile employment of meaningless jargon, have tended to suggest that the project is much further advanced than, in all probability, it really is. The fact that there appears to be no very high security rating attached to it in itself suggests that definite results have not yet been achieved; if and when they are, one would expect the usual "clamp" to be tightened down.

The *New York Herald-Tribune*, in a series of three articles printed in its issues for November 20, 21, and 22, 1955, stated that the following American organizations, among others, were involved:

The Glenn L. Martin Co. (Prime contractors, incidentally, of the Vanguard satellite vehicle and the Titan ICBM).

The Convair Division of the General Dynamics Corporation (Builders of the Atlas ICBM, and for whom Dr. Edward Teller, of the University of California, who is often described as the main creator of the fusion bomb, is top scientific consultant).

Lear, Inc. (a large electronics concern).

Clarke Electronics Labs.

The Princeton Institute for Advanced Study (Dr. Stanley Deser and Dr. Richard Arnowitt), and Princeton University itself.

The Universities of Indiana (Dr. Vaclav Hlavaty), North Carolina (Dr. Bryce S. de Witt), and Purdue.

A photograph was reproduced, showing Dr. Charles T. Dozier and Martin Kaplan, both of Convair, allegedly engaged on an anti-gravity experiment. They were shown facing some apparatus consisting of an assembly of vertical discs on a spindle, in casings supported on pillars, with what appeared to be electrical connections.

The Martin Co., which has actually run advertisements appealing for scientific research staff interested in gravity, are stated to have placed extramural contracts, through their Research Institute for Advanced Study (RIAS), with Dr. Pascual Jordan and Dr. Burkhard Heim, at the German Universities of Hamburg and Goettingen respectively.

Beyond this list, impressive enough in itself, other sources have linked the names of a number of other American firms and universities with the problem, and alleged that yet others in France, Italy, Japan, and elsewhere, are also involved. One school of thought has therefore concluded that something on the scale of the U.S. wartime "Manhattan District" programme, which produced the first nuclear fission bombs, is already in being. On the other hand, our own Arthur C. Clarke tells me that he recently discussed the matter with a well-known American science journalist in New York, and was assured that the whole business was a case of Much Ado About Nothing, started by "a bunch of engineers who don't know enough physics."

Clarke himself, incidentally, has an amusing theory to account for the start of the investigation at about this particular time; it is perhaps improbable, but, like all good stories, at least *deserves* to be true, and certainly to be repeated. We all know that his particular literary medium of science fiction is much more widely read by scientists, engineers and technical administrators than many of them care to admit . . . and it appears that the magazine *Astounding Science Fiction* published a story, about a couple of years ago, which had a striking relevance to all the current interest in "electro-gravitics." Called "Noise Level" and written by Raymond F. Jones, it described how the American authorities called together the nation's greatest scientific brains and demanded that they produce an anti-gravity device, claiming (quite untruthfully) that an unknown inventor had already done so, but that he had been killed in an accident while demonstrating his results, and all his records and apparatus destroyed. They supported this apocryphal story with some impressively-faked films and other convincing details, so that the scientists passed through a phase of scepticism into one of responding to the challenge with some intensive and inspired work. In due course, they came up with the required solution, in cruder and more cumbersome form than the device of their alleged anticipator—but anti-gravity, none the less. The moral of the story, of course, was the claim that nothing is impossible, once it is "known" to be practicable.

Now, says Clarke, is life deliberately imitating art? Did someone in the Pentagon read the story, and decide to see if some modification of this science-fictional plot could be transformed into science fact?

To return to the two extreme views mentioned earlier, it seems most unlikely that the more exuberant ("Manhattan District") one is justified; we can be fairly certain that it is unfounded, over-optimistic—to say the least, very premature. On the other hand, it is perhaps just as unlikely that the alternative, very pessimistic, extreme view is justified, either. It is much more probable that the work is, by modern standards, proceeding on a quite modest and exploratory scale, and that we simply have to wait and see how it will turn out. Much of it probably amounts to little more than the fact that American government and industrial agencies have begun to take an interest in some fundamental research work which was being done anyway, and to encourage and subsidize this, on the principle of casting the bread of dollar currency upon the waters of pure physics. A "guesstimate" which I have heard, that the total current expenditure is of the order of only 5 million dollars a year, is consistent with this view.

We can at least hope that, if the effort persists, the physicists in the long run may provide the essential new and basic knowledge for the "ignorant" engineers to use. It is an exciting thought that even this bare possibility exists: that before long, yet another scientific revolution (and one of the greatest of all time) might occur. It *might* even have started already, although (as we have said) this is rather unlikely. At any rate, there is one conclusion which can, even at this juncture, be stated with certainty:

If any anti-gravity device is ever to be developed, the first thing needed is a new discovery in fundamental physics—a new principle, not a new invention or application of known principles, is required.

The truth of this statement is clearly not realized by many of the journalistic writers on "electro-gravitics." The simple fact is, that at the moment, we do not know whether gravity nullification, reversal, or control of any kind, is possible or not. New fundamental knowledge might show that it is. On the other hand, it might place the whole concept in the same class of scientific absurdities as perpetual motion machines, or Wells' Time Machine for travel into our own established past; like them, it might violate some fundamental natural law of physics or logic. New knowledge of the sort required might come from theoretical researches in the more abstruse realms of mathematical physics, or from a more or less accidental experimental observation. Both these avenues have, in the past, led to fundamental scientific discoveries.

Devices for Overcoming Gravity

Also, a distinction should be drawn between a hypothetical device which opposes gravity in a fundamental sense, by the application of as yet unknown principles, and one which does so (or claims to do so) by the use of existing knowledge. This again is a difference not appreciated by many recent commentators.

Thus, several of them (e.g., *Interavia*, 1956, 11 (5), 373-4) have made somewhat vague and mysterious references to the work of inventors such as the American, Townsend T. Brown, who appear to be experimenting with model flying machines suspiciously like the legendary flying saucers, so-called "flying condenser discs" which generate lift by electrostatic charges. Between the two world wars, a British inventor named Verschoyle achieved some publicity with devices which, if they worked at all, did so by similar means. Nothing fundamental to the nature of gravity is involved, any more so than in the case of conventional flying machines, whether aerostats (balloons and airships), or aerodynes (aeroplanes, helicopters, etc.), which merely oppose gravity by other well-known forces. In all "aerodynes," for example, the lift (equal but opposite to the weight in level flight) is simply the aerodynamic reaction from the downward deflection, by wings or rotors, of a mass of air—literally a lifting jet (though not so obviously one as that of the Rolls-Royce "Flying Bedstead") and first-cousin to the exhaust that lifts a rocket.

The *New York Herald-Tribune* articles already quoted introduced, in places, similar confusions. They bracketed electrostatic lifting devices with the ion-rockets well-known to students of spaceflight, which are none the less rockets, even though they do generate their lifting or propulsive jets of charged particles electro-dynamically (by a voltage difference), instead of thermodynamically (by the temperature drop during expansion of a substantially neutral gas or smoke). Elsewhere, the same series ascribed undue significance (in this context) to the old trick phenomenon of "levitating" an object made of suitable paramagnetic material (e.g., aluminium) above the poles of an alternating-current electro-magnet. Here again, gravity is merely opposed, not in any way controlled—in this case by electromagnetic forces generated by eddy currents in the floating body.

All this is not to say that devices based on any of these electrical or magnetic phenomena might not conceivably be of some interest and practical importance

in their own right. The point to be made here is just that, even if they were, they would not represent anti-gravity, and would leave our power and influence over the basic gravitational forces exactly where it is today—in a position of complete ignorance and helplessness. We should merely have substituted one brute-force approach for another.

This would be equally true of another class of "anti-gravity" machine, if it were feasible at all, which it is not. This is the purely mechanical contraption of oscillating weights, or masses in non-uniform rotation: every few years, different inventors produce fresh versions of these, and it is sometimes difficult to spot their particular inherent fallacies (a few of them were discussed by the writer in a previous paper¹). However, fallacious they all are—if an upward lifting force is produced on one of their components, a compensating downward reaction will always exist on another. This is not always instantaneously true; that is, sometimes it is the net (upwards *minus* downwards) impulse which averages out at zero, over a complete cycle of operation.

If it did not, the inventor would have produced some sort of rocket, which might be defined as a device which opposes (but does not neutralize or control) gravity, by ejecting reaction mass to outside its own system. Once again, this is true of chemical, nuclear or ion rockets, and even (in the relativistic sense of equivalent mass) of Dr. Sanger's "photon rockets." None of these are closed mechanical systems, and no closed mechanical system could achieve a similar, gravity-defying, result.

H. G. Wells's famous Cavorite (of "The First Men in the Moon") is also fallacious, but for a different reason. If we had such a material as he described, and could use it as he envisaged, then we could defeat the law of the conservation of energy, and so achieve perpetual motion. However, this is not to say that, as far as we know at present, we could not build an anti-gravity device of some other sort, which worked by the continuous expenditure of appropriate quantities of energy (equally, as has already been pointed out, none of our present published knowledge suggests that we could). In passing, it is interesting to note that the nineteenth-century American astronomer, Simon Newcomb, now chiefly famous for his outspoken scepticism about the practicability of heavier-than-air flight by aerodynamic means, nevertheless went on record as believing that one day it would be achieved by the application of such yet (and still)-to-be-discovered principles.

Our Knowledge about Gravity

Gravity is, in fact, a most mysterious and intractable phenomenon. It is perhaps doubtful whether many people, even technically-trained people, realize just how true this is, or whether they notice the conspiracy of silence with which it is treated in most textbooks. One is almost reminded of a Polynesian "taboo," or the Victorian attitude to certain things, like sex or certain parts and functions of the body, considered "not quite nice." The student is taught that all masses attract one another according to the law: $F = \frac{Gm_1m_2}{d^2}$; he learns that it governs the stability of the Universe, and the various Newtonian equations which

describe its effects, such as $h = \frac{1}{2}gt^2$. If fairly sophisticated, he will proceed to learn the more accurate versions of the latter (such as: $h = \frac{2gR^2}{2gR - v^2} - R$), which must be used in celestial mechanics or for the calculation of spaceship orbits, to take account of the large variation in gravitational force over great distances, introduced by the inverse square law of the field. So he may become familiar with such concepts as escape-velocity, and may even be told something of Einstein's later relativistic modifications to Newtonian mechanics.

However, unless he is a very specialized post-graduate student in pure science, he is likely to be expected to accept the old idea of "action at a distance," and it is rather improbable that any of his teachers will draw his attention to our complete lack of knowledge of the physical relationship between gravitation and other phenomena, simply because we do know nothing, and the academic world apparently prefers not to advertise the fact! As the French mathematical physicist Borel wrote, many years ago now:

"There was, however, something rather strange in this phenomenon of gravitation, something that distinguished it from other physical phenomena. This was its utter immutability and its absolute independence of all external actions. Light is arrested by opaque bodies, deviated by prisms and lenses; electrical and magnetic actions are modified by the neighbourhood of certain bodies; gravitation alone remains always the same, and we have no means of enabling us to either increase or diminish it. Gravitation is indifferent to all physical circumstances, and is not affected by the chemical nature of bodies."

All this is just as true today as when he wrote it.

More recently, *Time* magazine quoted an interview with Einstein (April 10, 1950), in which the great man complained that physicists still acted as though gravitational effects did not exist; he said, "I do not believe that it is justifiable to ask, 'What would physics look like without gravitation?'" He himself, of course, firmly believed there was some relation between electromagnetism and gravitation, and spent the last years of his life working on a unified field theory to express it. He was anxious that his fellow-scientists should attempt to check it, but was quoted, in the same article, as saying himself: "Affirmation or refutation will not be easy, in spite of an abundance of known empirical facts. The derivation, from the equations, of conclusions which can be confronted with experience will require painstaking efforts and probably new mathematical methods."

Einstein was far from being the first scientist unwilling to accept gravitational effects without question, or without seeking to show their relationship to other, more tractable, natural forces. More than a hundred years ago, Michael Faraday was troubled by similar thoughts. Encouraged by his own classic successes in explaining the interdependence of chemical, electrical and magnetic phenomena, he conducted numerous experiments dropping weights down the old Waterloo shot tower on London's South Bank (by the present Royal Festival Hall), to try and discover some connection between gravity and magnetism or electrostatic charge. He was quite unsuccessful in these attempts; nevertheless, in a subsequent lecture he said: "They do not shake my strong feeling of the existence of a relation."

Sir William Crookes was another well-known physicist who worried about

this problem, and a search through the scientific journals of the last fifty years will reveal that many others kept him company. Although none of the work was successful, more was done than the meagre public reports might suggest. Thus, one encounters the names of Brush, Crémien, Austin, and Thuring; the Majorana brothers in Italy (who made important contributions to the development of the radio valve) once attracted quite a lot of attention with some measurements which seemed to show that gravitation was slightly absorbed by intervening substance, but these were later shown to be spurious.

A great modern physicist, Max Born, once wrote that perhaps when we had a more complete knowledge of the interaction of forces in the atomic nucleus we should find that gravitation was the result of something left over, a sort of incomplete compensation. I have seen a very similar expression of opinion by Prof. Marcus Oliphant, and in the articles from *The New York Herald-Tribune* already quoted, the following passage occurs:

"Dr. Deser and Dr. Arnowitt are of the opinion that very recently discovered nuclear and sub-nuclear particles of high energy, which are difficult to explain by any present-day theory, may prove to be the key that eventually unlocks the mystery. It is their suggestion that the new particles may prove to be basic gravitational energy which is being converted continually and automatically in an expanding universe 'directly into the most useful nuclear and electromagnetic forms.' In a recent scientific paper they point out: 'One of the most hopeful aspects of the problem is that until recently gravitation could be observed but not experimented on in any controlled fashion, while now with the advent in the past two years of the new high energy accelerators (the Cosmotron and the even more recent Berkeley Bevatron) the new particles which have been linked with the gravitational field can be examined and worked with at will.'"

If this point of view proves justified, then perhaps Einstein was mistaken when he deprecated the preoccupation of modern physicists with sub-nuclear particles, to the exclusion of the great cosmic force of gravity. Our best route to a solution may be by this very approach through the back door, as it were, then to scale up our knowledge from the sub-microscopic to the macroscopic. At any given time, there is always a region of intense scientific activity of which the layman, and the engineer, are almost entirely ignorant, even in the sense of being largely unaware of most of the things being discussed within it. Even with such an awareness, they would be almost incomprehensible, so perhaps it is as well that we do not hear about all the doubtful theories and experimental results, as yet unconfirmed, which are debated and disputed in the dim half-light of this region, like that which precedes the dawn of true understanding.

A fairly recent example of this sort, which did achieve some premature publicity, was Prof. P. M. S. Blackett's tentative law connecting the magnetic field of a body with its spin. Basing his suggestion on the striking coincidence of the value of the ratio of these quantities for the Earth, the Sun, and the star 78 Virginis (for which it had just been measured), Blackett argued, in 1947, that perhaps every spinning body generated a magnetic field in consequence. He deduced, from the three numerical values available, that perhaps $P = \frac{\beta G^{\frac{1}{2}}}{2c} \cdot U$,

where P is the magnetic field strength, β a new constant with magnitude near to unity for the units chosen, G the universal gravitational constant, c the speed of light, and U the angular momentum. Blackett said: "It is suggested

tentatively that . . . the above equation represents some new and fundamental property of rotating matter. Perhaps this relation will provide the long-sought connection between electromagnetic and gravitational phenomena."

Unfortunately, although a good deal of scientific interest and excitement was aroused at the time, this did not prove to be the case. The suggested law was apparently not supported by subsequent data, notably from the observation of certain of its deduced consequences, if it had been valid. So, what had appeared likely to open a crack in the door to gravitational knowledge proved, after all, to be just one more disappointment.

However, to the layman or technologist there now appear to be an increasing number of such stimulating speculations and possibilities. One leafs through an abstruse paper or an advanced textbook on theoretical physics, and, along with the neutrons, protons, electrons, positrons, neutrinos, mesons, and photons—all the ingredients of the Carrollian Wonderland of the nuclear physicist—one sometimes sees also the word "gravitons." Or one comes across a reference to "gravitational waves," or an argument about the speed of propagation of gravitational attraction; is it indeed the same as that of light, which is apparently what modern relativity theory would say? The suggestion has been made that certain experimental observations of the artificial satellite's orbit might provide some check on this point, and hence on current theories, and some anomalous observations, of the rate of expansion of the Universe.

Reference has already been made to the fact that Einstein, at the time of his death, was working on unified field theory—a sort of synthesis and extension of the relativity and quantum theories which would show the relationship between all kinds of physical phenomena, and perhaps explain them in terms of a sort of strain in space, consequent upon its true geometrical properties. The following quotation is from a broadcast talk given some years ago by Dr. Bronowski, on this subject of Einstein's last published work:

"One likeness between light, say, and gravity is that they both spread out into space in the same way, evenly in every direction from the point where they began; so that the strength of each falls off as the square of the distance from there. And then, of course, Einstein himself predicted in the theory of Relativity that light must be as it were attracted by matter, and was proved right spectacularly in recent eclipses. Another likeness has only become certain in the last few years, because it is another consequence of the theory of Relativity. This is the discovery that gravitation does not get everywhere instantly, but travels in waves like electro-magnetic waves, and at the same speed, which is the speed of light. Clerk Maxwell was sure that light and electro-magnetism are allied when his mathematics showed that they travel at the same speed. So it is at least tempting to couple gravitation with them now that we find that has the same speed too.

"Suppose you had asked Shakespeare in the year 1600 this question: 'What would happen on Earth if the Sun suddenly vanished into nothing?' He would have answered, 'It would get dark.' And if you pressed your question and asked, 'How soon?' Shakespeare would say, 'At once, of course.' No one in 1600 knew that it takes time for light to reach us from the Sun. And Shakespeare would not have given a thought to the path of the Earth if the Sun vanished. Because few people then believed that the Earth goes round the Sun, and those who did had no notion what kept it there.

"Now suppose you had put the same question to Christopher Wren 100 years later, in 1700. By this time it was known that it takes time for light to travel. And Christopher Wren, who was a friend of Newton's, of course also knew that it is the gravitational pull of the Sun which keeps the Earth on its path; but he supposed that it reaches everywhere instantly. So he would answer, 'If the Sun were suddenly to vanish into nothing now, the Earth would at once fly straight out of its orbit, but it wouldn't

get dark for about eight minutes.' And today? We know that gravitation travels no faster than light and other waves. So today we should say, 'If the Sun were suddenly to be snatched out of the sky, then the Earth would fly out of its orbit and it would get dark, both at the same moment . . . a little over eight minutes from now'."

Ever since I first heard Bronowski's talk seven years ago, I have thought this passage one of the most striking illustrations ever given of the growth of our scientific knowledge. Let us hope that its further growth will be stimulated by all the current official and industrial support apparently being given in America to those working, as Einstein was, on this important subject that Bronowski called "a unified theory which would at last forge the outstanding link between gravitation and electromagnetism." If "electro-gravitics" means anything, or is ever to do so, it must be just that.

Quite apart from the absorbing scientific questions involved, this last consideration raises another point of very great intrinsic interest. If the current American programme *does* succeed in producing any really significant result, it will be almost the first time in the history of science that a major fundamental advance has been obtained to direct order, primarily with a view to its application. As Sir Edward Appleton remarked in this year's Reith Lectures: "We cannot plan discovery; but we can plan the conditions in which discoveries can be made." The Americans would seem to be trying to go a little beyond this orthodox view.

The historical pattern has been almost invariably the same. A theoretician has deduced some fact, or even a general law, or else an experimenter has stumbled on it (often "stumbled" would be a rather ungenerous term, for the scientists would have been looking consciously in a region likely to be productive of valuable observations—which takes us back to Sir Edward Appleton). Further data are gained from special experiments designed to check up on the tentative theory, and so it is finally established. All this work has usually been the province of pure research workers at the Universities, although in recent years there has been a tendency for some fairly fundamental work to be carried out in the research laboratories of big industrial concerns. In any case, the motives of the scientists concerned have invariably been their private interests and curiosities, so that progress has been more or less spontaneous and sporadic.

At this stage, it has been rare in the past for any governmental or industrial agency to express any interest, other than possibly a philanthropic one, and certainly none whatsoever in any application of the new principles. Sometimes a few laymen (journalists or science-fiction writers!) do, but only to be scoffed at for their pains. Often the scientists concerned have been, at least in public, among the most sceptical; only rarely in the past have they been very interested in such a mundane matter as the application of their work, though this "ivory-tower" attitude is much less typical for the younger modern physicists.

The next stage has been for some pioneer technologist, or inventor, to apply the new principle in a novel device. Often he has had to face derision, opposition, and every sort of obstruction and frustration at first, but, finally successful, he (or all too often a later successor) finds his ideas adopted at last by the big interests of industry and government. A lot of people then say, of course, that

they had always realized this would happen, and it also becomes fashionable to point to the amazing foresight of the early lay prophets.

There have been signs for some time that changes are emerging in this classic pattern, for reasons which are fairly obvious. On the one hand, recent history has convinced the powers that be of the immense value of both pure research and technological development, and of the fact that the former, however academic it may appear at first sight, usually leads to the latter. On the other, modern technical advances are usually so complex and expensive that they are tending to be made *only* when the backing of big interests is available to foster them (University laboratories, for example, usually work now with big government grants, rather than private endowments).

Nevertheless, it has so far remained true that no really fundamental advance has been obtained to order; fundamental advances have been made with official backing in certain general areas of work, but that is not quite the same thing, though it may be approaching it. Nobody "ordered" the discovery of nuclear fission, for example, any more than that of radio-activity. Even in the field of invention, much the same considerations apply, although this question is a rather different one, and it is quite conceivable that even a major invention could be "ordered" to fulfil some specific need. However, Whittle's turbojet was born from his own spontaneous interest, in the traditional way—though radar was what the scientists produced in reply to an official request for something like a "death ray" to use in anti-aircraft defence.

Can the classic pattern, even as already modified by modern trends, be further completely transformed, by American methods of high-powered sponsorship? Faced with an urgent stated need and supplied with sufficient dollars, will the delicate flower of creative thought be so nurtured as to produce the necessary theories of brilliant insight, and to design the crucial revealing experiments? Or will these aids, which work all right for so-called "pump-handle" research, be in this case more like an excessive amount of chemical fertilizer, and kill the tender plant? Time alone will tell.

The Americans have said that they are worried by the relatively small amount of basic research undertaken in their country, and by the way in which their effort has tended, so far, to be deflected from this direction simply because of the enormous research subsidies now available from government and industry. Such agencies naturally tend to spend their money on something they already understand, to ask for things they know they would like to have. All this may be very relevant, for, even when trying to redress the balance by sponsoring some very novel and long-term basic research, what could be more understandable than that they should choose something, like the true nature of the gravitational field, which, if it could one day be manipulated, would have obvious practical applications?

The Martin Research Institute for Advanced Study (RIAS), which has said that it is interested in this subject, may be an expression of this line of thought. A brochure on its activities defines its guiding purpose as: ". . . to observe phenomena of nature and to encourage, promote and support investigations in search of underlying knowledge of these phenomena. To conduct theoretical and experimental studies, to discover the fundamental laws which affect them

and to evolve new technical concepts for the improvement and welfare of mankind." The freedom of action of the scientific staff is emphasized, and it is stated that: "The areas under investigation are not, in general, related to any immediate product development problems of the parent Company." Operating under the overall direction of Welcome W. Bender, RIAS itself describes its interest in "electro-gravitics" in these soberly-scientific terms:

"We are conducting an investigation of the general theory of relativity. Its aim is to understand the implications of the theory with regard to such things as the nature and meaning of singularities in the solutions to the field equations and the properties of gravitational radiation. Ultimately, it is expected that the programme will concern itself with generalizations of the theory by quantization, or otherwise, to see if such generalizations may have physical significance. At the same time, we are undertaking an experimental study of the effect of acceleration on the rates of natural clocks. This involves an investigation of spectral lines emitted by accelerating ions."

One final observation: if native Americans produce anti-gravity, by any methods, then the achievement will also be notable for being one of the first major scientific discoveries or inventions to their credit. This is not meant in any derogatory sense, for the Americans might well rest content with their great reputation for the subsequent development and application on a vast scale of the ideas of others. Nevertheless, it remains an historical fact (possibly an historical accident) that more original thought has tended, in the past, to emanate from Europe—together (let us admit) with a lot of short-sightedness about its application.

Let us, then, conclude by wishing our American cousins the greatest of good fortune with "electro-gravitics." If they succeed in producing an anti-gravity device, they will have revolutionized a good deal more than astronautics. Aeronautics, too, fairly obviously, but consider also the effects on the design and construction of buildings and other structures, the effect on the use of cranes, lifts, escalators, stairways—in fact, each and every kind of lifting device known to man. Very possibly, too, such a revolution would extend into almost all classes of machine, especially those producing, storing, or transforming power.

This publication, however, is concerned with astronautics, so let us return to the thought of the interesting problem awaiting Prof. Derek Lawden, or someone of his kind, in deriving all the equations of motion for a spaceship which could *reverse*, to any desired degree, the gravitational forces acting upon it from adjacent celestial bodies! And the rocket, after all the work and hopes lavished upon it for a couple of generations, might go the astronautical way of the dirigible in aeronautics—a vehicle which never fulfilled its promise.

REFERENCE

- (1) A. V. Cleaver, *J.B.I.S.*, 1947, 6, 127.

Index to the *Journal*, 1956, Vol. 15

It is regretted that pressure of other work has delayed the preparation of the index for Vol. 15 of the *Journal*. It is expected that it will prove possible to distribute the index with the July-September issue of the *Journal*.

THE BIEFELD-BROWN EFFECT

by
Carl Frederick Krafft

This interesting phenomenon consists of the tendency of a charged electric condenser to move in the direction of the positive pole, despite the principle of action and reaction.

In an electric condenser, just as in a hydrogen atom, there is a direct flow of ether from the positive charge to the negative charge. Since the ether is incompressible, its streamlines cannot come to abrupt ends within the charges. The positive charge must therefore take the ether in from the surroundings and the negative charge must deliver it to the surroundings. This will necessarily diminish the external ether pressure on the positive side and increase it on the negative side.

If we adopt the modern view that the force of gravity is due to a pressure gradient in the ether, then such a difference of ether pressure on the two sides of a condenser should force the entire condenser in the direction of the side that carries the positive charge, exactly as has been observed experimentally. A charged condenser should therefore weigh more when its positive side faces down than when it faces up.

(Reprinted from BSRF's ROUND ROBIN, Vol. XII, No. 3, Sept-Oct 1957)

J. Borderland Research

Mar-Apr 1987 JBR, Page 11

43 (2) March-April
87

THE FLYING SAUCER

A SIMPLIFIED EXPLANATION OF THE APPLICATION
OF THE BIEFELD-BROWN EFFECT TO THE SOLUTION
OF THE PROBLEMS OF SPACE NAVIGATION

by

Mason Rose, Ph.D., President
University for Social Research

← "DEFUNCT" ~~(scribbled out)~~

The scientist and layman alike encounter a primary difficulty in understanding the Biefeld-Brown effect and its relation to the solution of the flying saucer mystery.

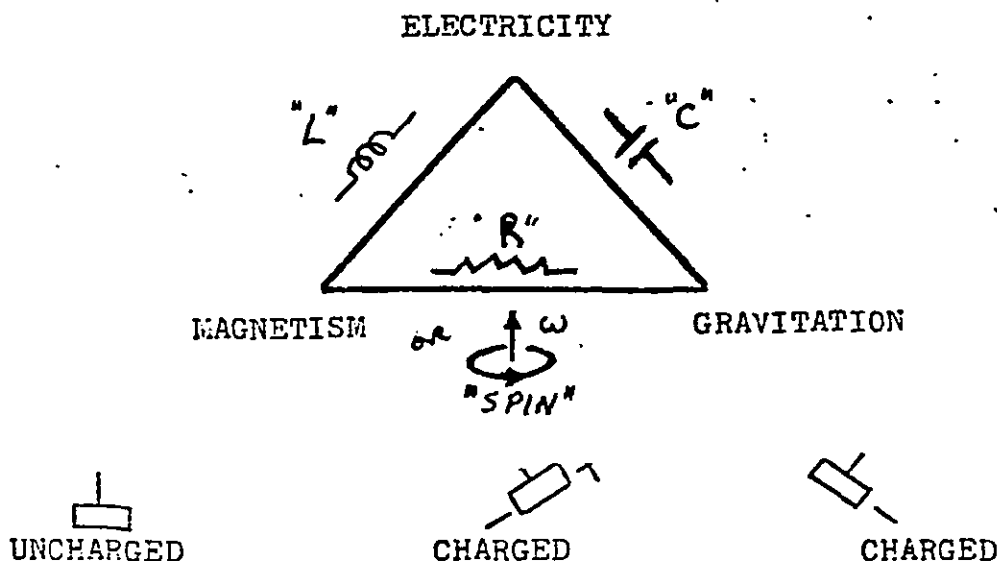
This difficulty lies in the fact that scientist and layman alike think in electromagnetic concepts, whereas the Biefeld-Brown effect relates to electrogravitation.

Neither scientist nor layman can be expected to know the details of electrogravitation, inasmuch as it is a comparatively recent and unpublished development. Townsend Brown is the discoverer of electrogravitational coupling.

To date, Townsend Brown is the only known experimental scientist in this new area of scientific development. Thus anyone who wishes to understand electrogravitation and its application to astronautics must be prepared to lay aside the commonly known principles of electromagnetics in order to grasp the essentially different principles of electrogravitation. Electrogravitational effects do not obey the known principles of electromagnetism. Electrogravitation must be understood as an entirely new field of scientific investigation and technical development.

Perhaps the most efficient method of inducing an understanding of electrogravitation is to review the evolutionary development of electromagnetism.

From the smallest atom to the largest galaxy, the universe operates on three basic forces, namely, Electricity, Magnetism and Gravitation. These three forces can be represented as follows:



Taken separately, neither is of much practical use. Electricity by itself is static electricity and therefore functionless. It will make your hair stand on end, but that is about all.

Magnetism by itself has very few practical applications aside from the magnetic compass, whereas gravity simply keeps objects and people pinned to the earth.

However, when these are coupled to work in combination with each other, almost endless technical applications arise. To date, our total electrical development is based on the coupling of electricity with magnetism, which provides the basis for the countless uses we make of electricity in modern societies.

Farraday conducted the first productive empirical experiments with electromagnetism around 1830, and Maxwell did the basic theoretical work in 1865.

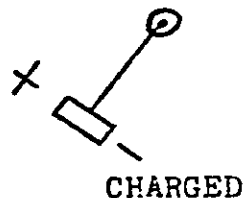
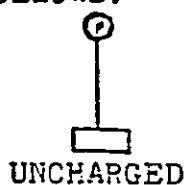
The application of electromagnetism to microscopic and sub-microscopic particles was accomplished by Max Planck's work in quantum physics about 1890; and then in 1905 Einstein came forward with relativity, which dealt with gravitation as applied to celestial bodies and universal mechanics.

It is principally out of the work of these four great scientists that our electrical developments, ranging from the simple light bulb to the complexities of nuclear physics, have emerged.

In 1923, Professor Biefeld of Dennison University suggested to his protege, Townsend Brown, certain experiments which led to the discovery of the Biefeld-Brown effect and ultimately, to the electrogravitational energy spectrum. After 28 years of investigation by Brown into this coupling effect between electricity and gravitation, it appears that for each electromagnetic phenomenon there exists an electro-gravitational analogue. This means, from the technical and commercial viewpoint, potentialities for future development and exploitation as great or greater than the present electrical industry. When one considers that electromagnetism is basic to the telephone, telegraph, radio, television, radar, electric generators and motors, power production and distribution, and is an indispensable adjunct to transportation of all kinds, one can see that the possibility of a parallel, but different, development in electrogravitation has almost unlimited prospects.

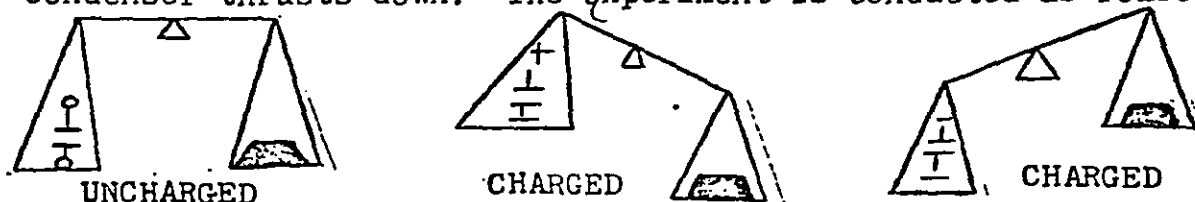
The first empirical experiments conducted by Townsend Brown had the characteristic of simplicity which has marked most other great scientific advancements. These concerned the behavior of a condenser when charged with electricity.

The first startling revelation was that if placed in free suspension with the poles horizontal, the condenser, when charged, exhibited a forward thrust toward the positive pole. A reversal of polarity caused a reversal of the direction of the thrust. The experiment was set up as follows:



The antigravity effect of vertical thrust is demonstrated by balancing a condenser on a beam balance and then charging it. After charging, if the positive pole is pointed upward, the condenser moves up.

If the charge is reversed and the positive pole pointed downward; the condenser thrusts down. The experiment is conducted as follows:



These two simple experiments demonstrate what is now known as the Biefeld-Brown effect. It is the first and, to the best of our knowledge the only method of affecting a gravitational field by electrical means. It contains the seeds of control of gravity by man. The intensity of the effect is determined by five factors, which are:

1. The separation of the plates of the condenser, the closer the plate the greater the effect.
2. The ability of the material between the plates to store electrical energy in the form of elastic stress. A measure of this ability is called the "K" of the material. The higher the "K", the greater the Biefeld-Brown effect.
3. The area of the plates, the greater area giving the greater effect.
4. The voltage difference between the plates, more voltage, more effect.
5. The mass of the material between the plates, the greater the mass, the greater the effect.

* It is this fifth point which is inexplicable from the electromagnetic viewpoint and which provides the connection with gravitation.

On the basis of further experimental work from 1923 to 1926, Townsend Brown in 1926, described what he called a "space car". This was a revolutionary method of terrestrial and extra-terrestrial flight, presented for experiment while motor-propelled planes were yet in a primitive stage.

This engineering feat by Townsend Brown, was ~~all~~ the more remarkable when we consider such a machine produces thrust with no moving parts, does not use any aerodynamic principles of flight, and has neither control surfaces, nor a propeller. Townsend Brown had discovered the secret of how the flying saucers fly, years before any such objects were reported.

Now that the basic difference between electromagnetism and electro-gravity has been indicated and the basic principles of the Biefeld-Brown effect have been outlined, we are finally ready to understand the principles of astronautics or the conquest of space.

* - ϵ , "ELECTROGRAVITIC FORCE" $\propto M \left(\frac{\epsilon \cdot A}{d} \right) \propto MC$! [where "C" = CAPACITANCE in "FARADS"]

The earth creates and is surrounded with a gravitational field which approaches zero as we go far into space. This field presses objects and people to the earth's surface; hence it presses a saucer object to the earth.

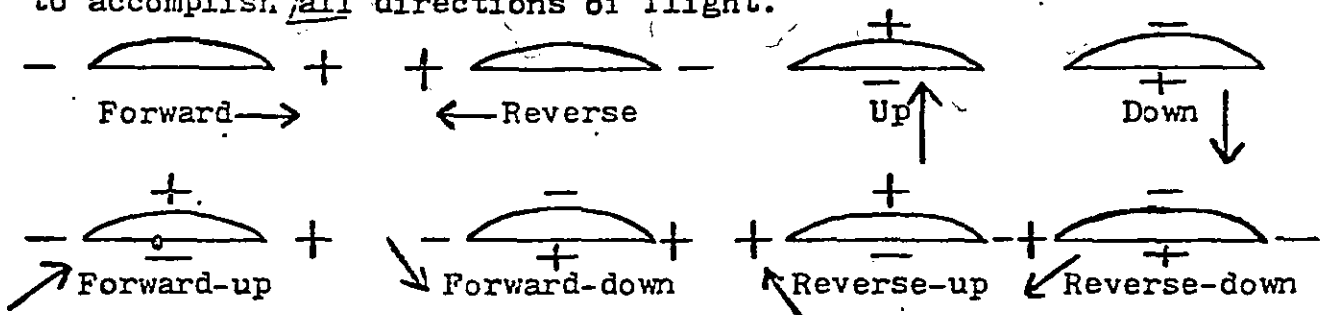
However, through the utilization of the Biefeld-Brown effect, the flying saucer can generate an electrogravitational field of its own which modifies the earth's field.

5a2
P.9

This field acts like a wave, with the negative pole at the top of the wave and the positive pole at the bottom, the saucer travels like a surfboard on the incline of a wave that is kept continually moving by the saucer's electrogravitational generator.

Since the orientation of the field can be controlled, the saucer can thus travel on its own continuously generated wave in any desired angle or direction of flight.

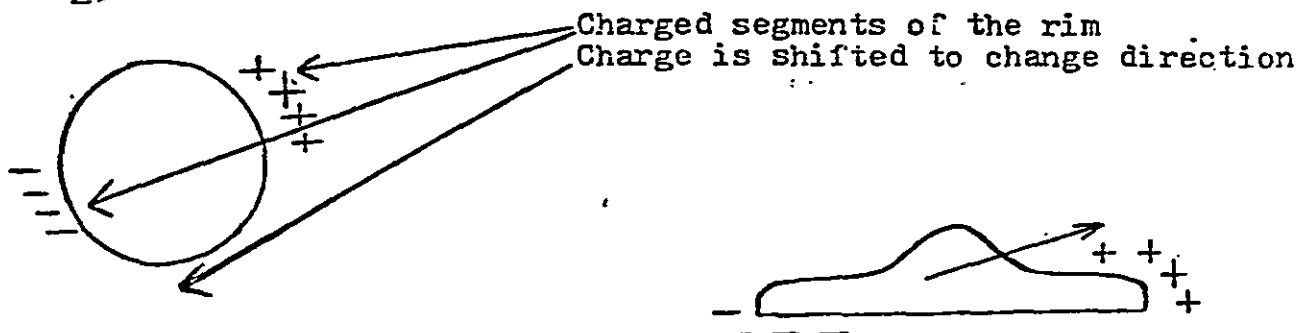
The method of controlling the flight of the saucer is illustrated by the following simple diagrams showing the charge variations necessary to accomplish all directions of flight.



Since the saucer always moves toward its positive pole, the control of the saucer is accomplished by varying the orientation of the positive charge. Control, therefore, is gained by switching charges rather than by control surfaces. Since the saucer is travelling on the incline of a continually moving wave which it generates to modify the earth's gravitational field, no mechanical propulsion is necessary.

Once we understand that the horizontal and vertical controls are obtained by shifting the positive pole which turns the field, then we are in a position to extrapolate a finished saucer design.

The top view would be as follows:



Upper plate charged positive, lower negative, for lift, resultant direction between thrust and lift indicated by arrow.

The saucer's edge would contain a number of conductor segments, and the saucer would turn in any direction simply by shifting the positive and negative charges to appropriate positions along its edge.

The vertical thrust would be regulated by varying the positive charge on top of the saucer, the amount of thrust being regulated by the amount of charge generated.

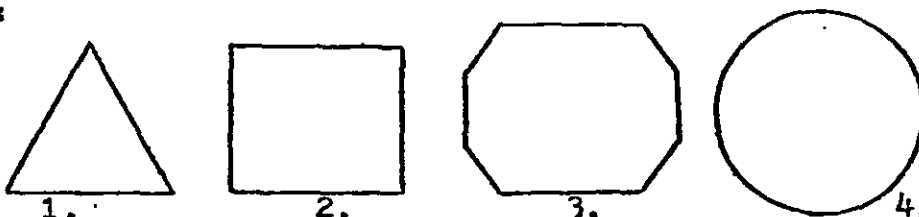
Flying saucers in all probability do not utilize external controls for direction, nor do they have any visible means of propulsion. The flying saucers fly on an entirely new principle, namely, the Biefeld-Brown electrogravitational effect, and hence do not utilize any of the standard aerodynamic principles of an airfoil. Flying saucers cannot be understood from the traditional principles of aeronautical engineering.

To understand the flying saucers an individual must temporarily bypass these points of view, to learn about a new principle, The Biefeld-Brown electrogravitational effect, and then return to the older points of view for critical, theoretical analysis and empirical testing.

Years ago, long before saucers as such were reported by observance, Townsend Brown developed a captive flying saucer, that is, a scale model saucer with a free bearing going around a stationary pole.

Brown did not start with round objects, in fact, the first object that flew, was a triangle (1), the next a square (2), then a square with the edges cut off (3), and finally a round shaped saucer (4).

The evolutionary development could be graphically expressed as follows:



Since experiments proved the saucer shape most effective, the changes were made for empirical reasons.

Having solved the problem of horizontal thrust, Townsend Brown developed a profile shape which would be most efficient to shape the electrogravitational field for maximum vertical thrust. The final profile that developed was the shape illustrated here:



The first report of a disc shaped object in the sky dates back to the sixteenth century. At long intervals during the centuries since have come other reports. Most of them are undoubtedly unreliable as observations, distorted by telling and retelling. But in these older reports, as well as in the very numerous series which has accumulated since 1947, there is a teasing common thread concerning appearance and behavior which makes any certainties about the unreality of flying saucers very insecure.

One of the great difficulties in substantiation of these reports is that in both appearance and behavior, these objects seem to be simple, scientific impossibilities. Here are some of the reasons advanced by technical men to prove the impossibility of devices such as the reports describe:

1. The reports reveal, in most cases, no method of propulsion which can be understood. There are no propellers in any of the reports. Some of the reports describe a long flame jet trailing behind a cigar-shaped object. But this flame is orange red in color, indicating an inefficient combustion which would make it ineffective as a reaction jet such as propels rockets and jet planes. No other known physical laws seemed capable of explaining the observed motion of the objects.

2. The reports describe a range of speed and acceleration from stationary hovering to speeds greater than present day rockets can deliver, and the changes of rate of motion, the accelerations, are far beyond the capacities of any known man-made vehicles. Flight experts point out that such accelerations would impose impossible stresses on any human or human-like occupants. Therefore, they say, the reports must be false or erroneous.

3. Many of the reports concern night sightings and describe a glow, usually of blue or violet color, around a periphery of the objects. Physicists have noted that such a glow is characteristic of a very high voltage electrical discharge, but add that this suggests no means of explaining the appearance or behavior of the objects described in the reports.

4. The description of shapes and performance seems to indicate a complete or almost complete disregard of aerodynamic principles. The objects seem not to need the support of air as a plane does, nor to depend on the lift provided by properly designed surfaces moved rapidly through the air medium.

These are weighty arguments, PROVIDED THE ASSUMPTIONS BEHIND THEM ARE CORRECT. But now comes physicist Townsend Brown, who has spent the last 28 years exploring the consequences of a simple experiment he performed at the suggestion of Dr. Biefeld in 1923. Dr. Biefeld, professor of physics and astronomy at Dennison University, former classmate of Einstein in Switzerland, wondered if an electrical condenser, hung by a thread, would have any tendency to move when it was given a heavy electrical charge. Townsend Brown provided the answer. There is such a tendency. But the attempt to understand and explain this motion has occupied him ever since, and led to discoveries of truly basic importance.

The observed motion of a charged condenser has been labeled the Biefeld-Brown effect. Studying this effect, Brown pointed out in 1923 that this tendency of a charged condenser to move might easily grow into a new and basically different method of propulsion.

By 1926 he had described a "space-car" utilizing this new principle. By 1928 he had built working models of a boat propelled in this manner. * By 1938 he had shown that his specially designed condensers not only moved, but had certain interesting effects on plants and animals.

All of this, while very exciting, is for most of us just a repetition of the story scientific development so characteristic of our age. But now comes the unexpected. Townsend Brown, working in his laboratory, building models and trying endless variations of size, shape and design of his charged condensers, made a flying saucer which flew around a maypole, BEFORE FLYING SAUCERS BECAME A NEWSPAPER TOPIC. AND THE REASONS LISTED ABOVE, WHICH LED THE SPECIALISTS TO REJECT THE REPORTS ON OBSERVED SAUCERS, PROVED TO BE BOTH EXPLICABLE AND NECESSARY TO THEIR OPERATION UNDER THE ELECTROGRAVITATIONAL PRINCIPLE.

Let us look at the four main objections in a new light:

1. No understood method of propulsion. The saucers made by Brown have no propellers, no jets, no moving parts at all. They create a modification of the gravitational field around themselves, which is analogous to putting them on the incline of a hill. They act like a surfboard on a wave. The surfboard moves without propellers or jets too. But it is confined to the direction and speed of the water wave. The electro-gravitational saucer creates its own "hill", which is a local distortion of the gravitational field, then it takes this "hill" with it in any chosen direction and at any rate.

2. The second objection concerned the tremendous accelerations which, on the basis of previous technology, would subject any animal occupants to unbearable stresses. But, says Brown, the occupants of one of his saucers would feel no stress at all, no matter how sharp the turn or how ~~great~~ the acceleration. This is because the ship and the occupants and the load are all responding equally to the wavelike distortion of the local gravitational field. In an airplane the propeller pumps air backward and by reaction itself moves forward. The reaction thrust on the propeller is transferred to the frame of the aircraft. This frame then shoves the load and occupants forward CONTRARY TO THEIR NATURAL TENDENCY TO MOVE AT A CONSTANT RATE IN A CONSTANT DIRECTION. But in the saucer ~~no~~ such transfers of thrust from one member to another occurs. The entire assembly moves in unison in response to the locally modified gravitational field. The nearest analogy in our experience is going down in an elevator. When the elevator starts down, it is not necessary for the elevator to shove on our bodies, both elevator and passengers share a gravitational tendency to move down. They do so without any shoving or any stresses between elevator and passengers.

3. Townsend Brown's saucers require a highly charged leading edge, the positive pole. But such a charged pole produces an electrical corona. In the largest models made, this develops a decided bluish-violet glow easily visible in darkness or a dim light. A full scale ship operating on this principle would be expected to produce a spectacular corona effect visible for many miles.

4. The outlines and shape of Brown's saucers were the result of electrogravitational considerations, not the result of wind tunnel tests of aerodynamic designs.

* - Where IS all this Research NOW?

For they move, not on the lift of air, but on the lift of a modified gravitational field. In operating saucers such aerodynamic considerations would have to be taken into account to reduce drag and friction, but not to produce lift and thrust.

5. And, finally, when Brown turned his attention to improved ways of generating high voltages, the most promising new method involved the use of a flame jet to convey negative charges astern. This flame was relatively inefficient as a generator if it was adjusted for the best combustion of the fuel. But if it was adjusted to an orange-red color, indicating incomplete combustion of fuel, it conveyed the charges very effectively and set up the required negative space charge behind the ship.

The reasons advanced by the experts to "explain away" the saucer reports, when seen from a new and different viewpoint, appear to be the specific reasons why they ~~can~~ operate, on electrogravitational rather than electromagnetic principles.

The next opinion which must be corrected is the idea of overly intensified supersonic vibration. The Townsend Brown experiments indicate that the positive field which is travelling in front of the saucer acts as a buffer wing which starts moving the air out of the way. This immaterial electrogravitational field acts as an entering wedge which softens the supersonic barrier, thus allowing the material/leading edge of the saucer to enter into a softened pressure area. Diagrammed, this would be illustrated as follows:



The University for Social Research is ready to offer this experimental finding to the jet airplane and guided missile industry as a practical method of softening the supersonic barrier.

It should be noted that in a jet plane or guided missile the extra weight added to create the Biefeld-Brown electrogravitational effect would be compensated for by the added thrust created by the movement of the plane toward the positive field created in front of the leading edge.

As we have previously stated, for every known electromagnetic effect there is an analagous electrogravitational effect but electrogravitational application and results differ from those of electromagnetic. (This presupposes that an entirely new electrogravitational industry comparable to the present electromagnetic industry will emerge from the theoretical formulations and empirical experiments of Townsend Brown.

The University for Social Research, in presenting the Biefeld-Brown electrogravitational effect, offers to the world new vistas of increased production, betterment of human living and additional economic stability to all countries.

Mason Rose, Ph. D., President
University for Social Research.

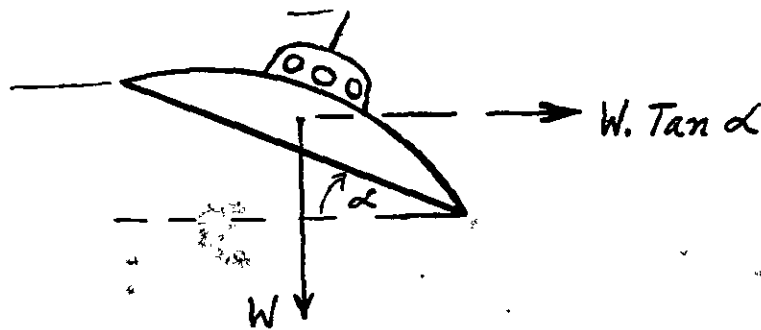
April 8, 1952.

* - SAME "colour" as "SIDEWISE-FLYING SPINDLE" (U.F.O.), Seen by ME!
T.T. Brown spent 2 YEARS as "CONSULTANT" for FRENCH "MINISTRY of Aircraft (!) CONSTRUCTION"
T. Brown's RESEARCHES are available from WILLIAM L. MOORE, 4219 W. OLIVE (SUITE 247),
BURBANK, CALIF., 91505, U.S.A.

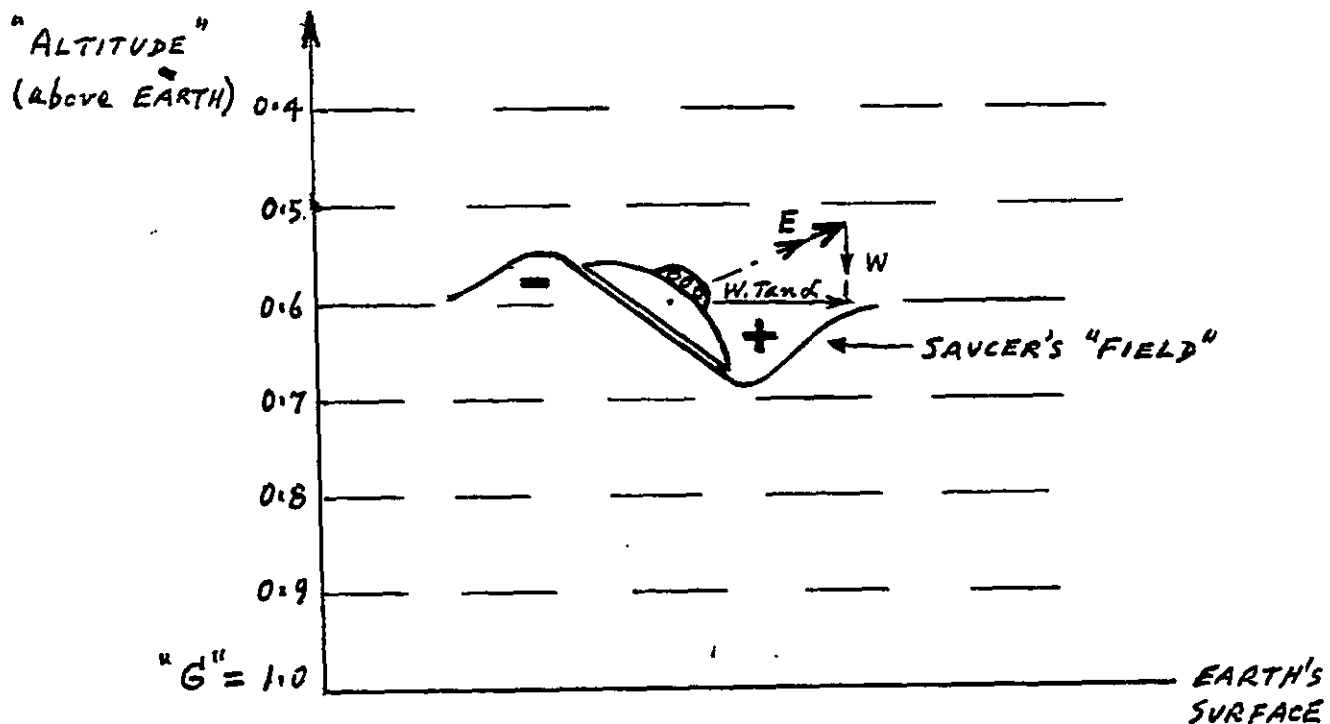
- 9 -

BILL WHAMOND
Montreal

- APPENDIX -



"FLIGHT-PHYSICS"; — is just the "HIGHSCHOOL-PHYSICS" of a "RAMP".



FLYING-Saucer "SURFBOARDING" along one of
EARTH'S "GRAVITATIONAL EQUIPOTENTIAL-LINES".
(E is the "ELECTRIC FIELD-STRENGTH" VECTOR).

" GRAVITY IS A PUSH "

W.C. WRIGHT

Comets have long puzzled the science community because as comets approach heavenly bodies they are deflected around these heavenly bodies that are in their paths..... Isaac Newton stated back in the 17th Century that the ATTRACTION between MASSSES would be GREATER the CLOSER masses became in regard to one another. This physical model might demonstrate why the theory of gravity being an ATTRACTION has puzzled the science community. If you apply Wright's theory of gravity being a magnetic repulsion between all heavenly bodies, then you won't be puzzled.

On October 19, 1980, Dr. Carl Sagan presented his 4th segment of his \$8,000,000 production called COSEOS. He said the odds for a comet to hit another body in space is 1 billion to 1 and he demonstrated this with animated drawings, but nothing physical. Don't you find that a very odd way to demonstrate this space action?

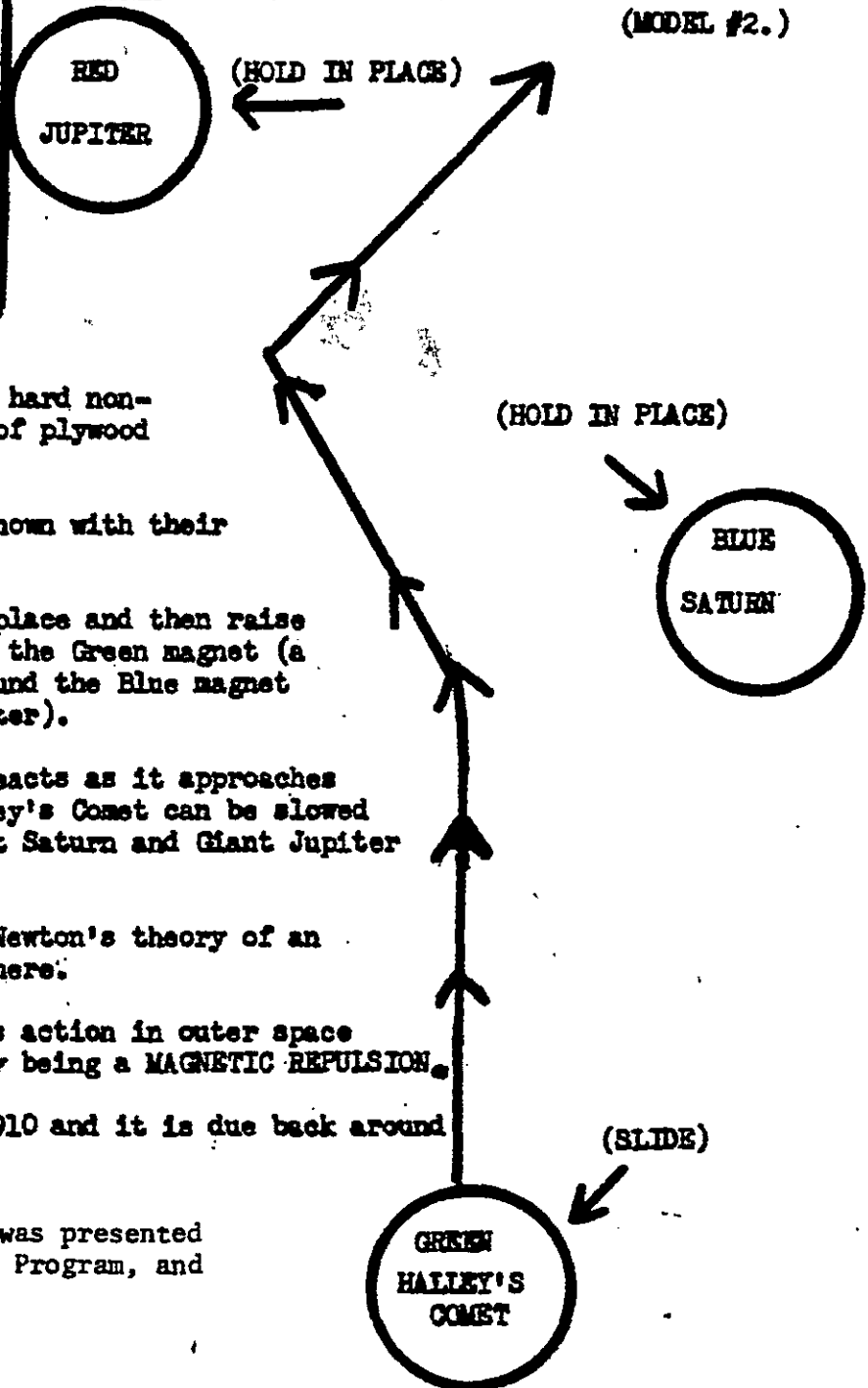
(MODEL #2.)

INSTRUCTIONS HOW TO OPERATE MODEL:

1. Place this experiment on top of a hard non-magnetic surface such as a piece of plywood or a large hardbound book.
2. Place the magnets on the colors shown with their painted sides up.
3. Hold the Blue and Red magnets in place and then raise the plywood or hardbound book, so the Green magnet (a comet) will slide and deflect around the Blue magnet (Saturn) and the Red magnet (Jupiter).
4. You just saw how Halley's Comet reacts as it approaches our planets every 76 years. Halley's Comet can be slowed down as much as 15 months if Giant Saturn and Giant Jupiter are in its way.
5. IT'S PUZZLING to understand how Newton's theory of an ATTRACTION OF MASSSES would apply here.
6. It is very easy to understand this action in outer space if you apply the theory of gravity being a MAGNETIC REPULSION.
7. Halley's Comet last appeared in 1910 and it is due back around 1986.

(On November 18, 1980 this theory was presented in the Ottawa University Extension Program, and was received very favorably.)

(RAISE THE PLYWOOD OR HARDBOUND BOOK HERE.)



—UNIVERSAL GRAVITY COULD BE A MAGNETIC PUSH—

The North Poles on these magnets are painted Red, Green and Blue.

These experiments are so designed that the LIKE POLES on all three magnets oppose each other and the basic law of magnetism states—"LIKE MAGNETIC POLES REPEL".

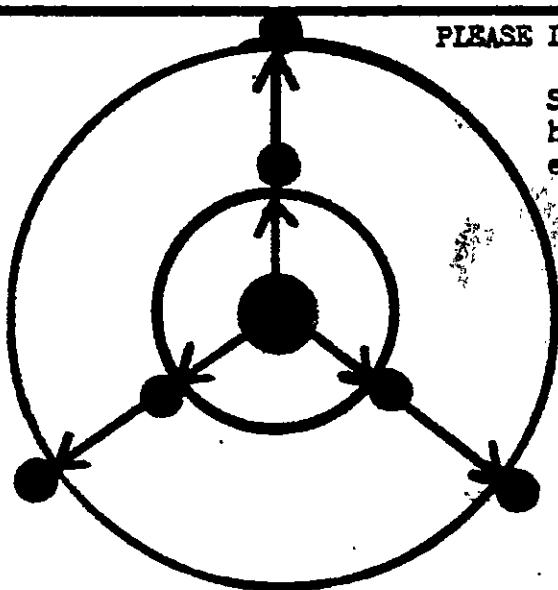
You will find these experiments, regardless of your age, very fascinating, interesting and educational. They should not be considered a toy. The Wright Gravity Theory physically provides evidence that gravity could be a PUSH between all heavenly bodies.

PLEASE LET AT LEAST 2 STUDENTS IN PHYSICS WORK THESE MODELS.

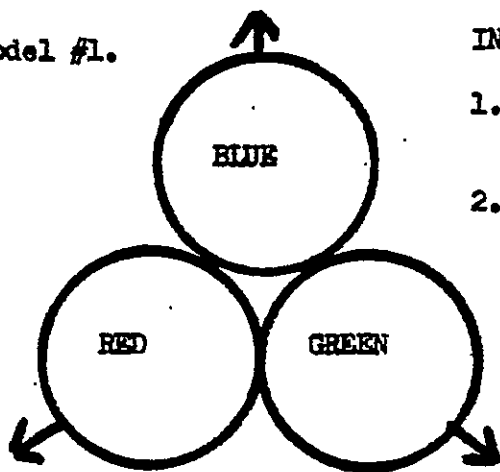
Similar sketches can be found in science books depicting how our universe is expanding faster and faster each and every second in rings based on the BIG BANG THEORY.

MAUPIN'S INSURANCE COMPANY IS LOCATED AT 1652 WEST TEXAS ST., SUITE 102, FAIRFIELD, CALIFORNIA.

THESE MAGNETS CAN BE PURCHASED AT RADIO SHACK LOCATED IN FAIRFIELD AT 2449 NORTH TEXAS ST.



Model #1.



INSTRUCTIONS HOW TO OPERATE THIS BIG BANG MODEL.

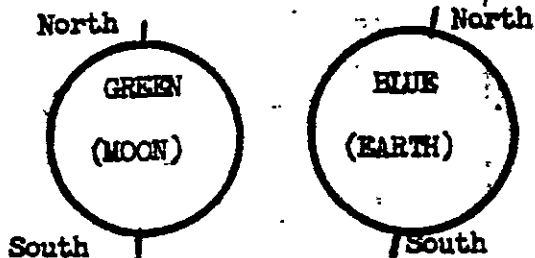
1. HOLD the magnets on the colors shown with their painted sides up.
2. Release all three magnets at the same time and they will expand or PUSH apart very rapidly.

On September 28, 1980, Dr. Carl Sagan presented the first segment of his \$8,000,000 production called "COSMOS". He showed the Big Bang Theory with flashing colored lights but no physical working models. Doesn't that strike you as being very odd? Wright's Big Bang Model shown here cost about 25 cents and it does provide physical evidence to support his theory that gravity could be a magnetic repelling factor.

On October 11, 1980, Wright sent this model to Dr. Carl Sagan for his evaluation.

A full set (4 experiments) of these models: DONATION- 25 cents. They will be mailed to you via First Class Mail for only \$1.00. Address is W.C. WRIGHT—732 Ohio St., Fairfield, Calif. 94533... (707) 429-0598. My hardbound book called GRAVITY IS A PUSH sells for \$7.00.

(Magnets have always been very fascinating to most people. These magnets when purchased cost about 15 cents apiece. They are cheaper in lots of 100. You are getting these models and magnets for only 25 cents due to MAUPIN'S INSURANCE CO. AND RADIO SHACK SHOWN ABOVE. I AM VERY GRATEFUL FOR THEIR INTEREST IN THE THEORY. W.C. Wright)



A similar sketch can be found in a book called "Geology of the Moon" by Mutch.

The Earth and the Moon are both large magnets with their North and South Poles opposing each other.

Science books state that "LIKE POLES REPEL".

HOW TO OPERATE THIS MODEL.

1. Place the Green Magnet (Moon) on color shown.
2. Place the Blue Magnet (EARTH) on color shown.
3. Move the Green Magnet (Moon) towards the Blue Magnet (Earth) and the Blue Magnet will be repelled away. (This sketch provides evidence to support the Wright theory that states that gravity could be a magnetic repulsion between all bodies in space.

"The Greatest Scientist Of The 20th Century", Einstein, back in the early 1900^s, said that heavenly bodies do not "pull" each other and objects are "pushed down" to Earth. He tried to write a set of equations that would unite gravity and electro-magnetism but he failed.

oOo

In 1968 I said heavenly bodies "push" each other and objects are "pushed down" to earth and gravity could be magnetic. I was not aware of what Einstein had said years before. In 1976 I humbly feel I wrote a set of equations that do unite gravity & electro-magnetism.

oOo

I will now show some of my formulas.

$$\text{mps} = \frac{188}{\text{kn} \sqrt{r}} \quad r = \left(\frac{188}{\text{kn} \text{ mps}} \right)^2$$

(Example—"r" for Earth is 93 mega-miles from Sun.)

Symbols

r = mega-miles from orbiting body.
mps = miles per second in orbit.
kn = key number.
188 = a basic number.

These formulas are electronic oriented and that is why I was able to write them because I felt electro-magnetism could be related to gravity. (This was Einstein's belief but failed.)

oOo

The key number for the planets that orbit the sun is 1.06.
The key number for the satellites that orbit Mars is 2,350.
The key number for the satellites that orbit Jupiter is 34.2.
The key number for the satellites that orbit Saturn is 62.25.
The key number for the satellites that orbit Uranus is 158.
The key number for the satellites that orbit Neptune is 147.
The key number for our moon that orbits Earth is 618.

W.C.Wright—732 Ohio St.
Fairfield, Ca. 94533
(707) 429-0598. I have a
25 minute slide-sound movie
if you would care to see it.
My hardbound autographed book
GRAVITY IS A PUSH is \$7.00

I feel this key number can be applied to any body that is being orbited in the Universe.



Largely self-taught, Robert T. Jones of NASA's Ames Research Center became one of the world's leading aeronautical designers. Among his contributions are the principles of the swept wing and pivoting wing.

(Did you know Einstein was a self-taught drop-out?)

Wright is like Einstein and R.T.Jones, because he is self-taught, too. The greatest thing about a self-taught person is that they only have one limitation, and that is their imagination.

R.T.Jones is living proof of that.

W.C. Wright for

Hybrid Electrostatic/Acoustic Levitator

Particles and drops can be rotated and vibrated.

NASA's Jet Propulsion Laboratory, Pasadena, California

A levitator for liquid drops and small particles includes both electrostatic and acoustic components for versatility in the manipulation of samples. The sample is levitated electrostatically and rotated or vi-

brated acoustically. Because the electrostatic and acoustic forces are independent of each other, the hybrid levitator is especially suitable for studies of drop dynamics. Like all-acoustic or all-electrostatic

systems, it can also be used in studies of containerless material processing.

The levitator (see Figure 1) includes two dish-shaped electrodes that generate the levitating electrostatic field. A liquid-

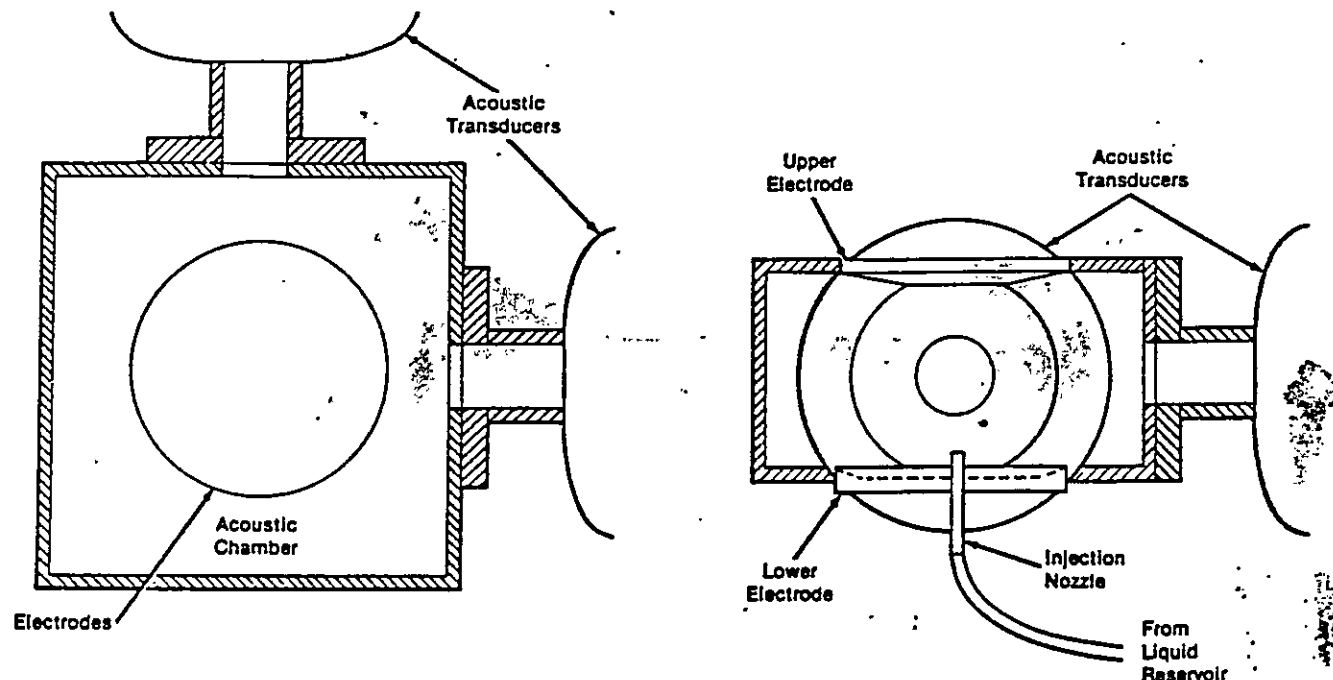


Figure 1. The Vertical Levitating Force is applied to the sample by the upper and lower electrodes. Torques or vibrational forces in the horizontal plane are applied by the acoustic transducers.

injection nozzle enters the levitation chamber through a hole in the center of the lower electrode. An acoustic transducer is attached to each of two orthogonal vertical faces to produce orthogonal acoustic standing waves in the chamber.

To facilitate the adjustment and stabilization of the vertical position of the sample, the electrostatic portion of the levitator is controlled by a digital feedback system. The acoustic transducers are locked to each other in frequency and phase. To apply an acoustic torque to rotate the sample, a phase difference of 90° is imposed. The torque is varied by adjusting the amplitude of the sound.

A nonrotating sample can be made to vibrate by fixing the phase difference at zero and sinusoidally modulating the amplitude. Both a nonzero phase difference and amplitude modulation can be used to produce rotation and vibration. If the sample is a liquid drop, it can be rotated until it breaks in two (see Figure 2). The levitator can also manipulate aluminum shells, glass shells, and wooden spheres.

This work was done by Won K. Rhim, Eugene H. Trinh, Sang K. Chung, and Daniel D. Ellerman of Caltech for NASA's Jet Propulsion Laboratory. NPO-16834

Figure 2. An Electrically Charged Water Drop about 4 mm in diameter was levitated electrostatically and rotated acoustically until it assumed a dumbbell shape and broke apart.

énergétique des faisceaux dans les grands accélérateurs actuellement en service.

Une preuve expérimentale particulière de ce phénomène peut être aisément donnée en concentrant, sur une cible solide, le faisceau infrarouge d'un laser de puissance. Les effets constatés sont, sans aucun doute, le fait de particules ayant, dans le milieu d'observation, une énergie bien supérieure à celle qu'un photon infrarouge isolé serait en mesure de fournir.

matérielle et
gravitationnelle

R.L. Vallée

11

ÉLECTROGRAVITATION ET PROPULSION ÉLECTROMAGNÉTIQUE

Nous avons vu qu'une onde électromagnétique était accompagnée d'un champ de gravitation égal à $\gamma = \frac{1}{\rho} \cdot \frac{\partial}{\partial t} (E \wedge H)$. Ce champ doit être, en général, extrêmement faible. Bien que nous en ignorions la valeur exacte, ρ représente sans doute une énergie très élevée.

Afin de pouvoir évaluer les possibilités que peut offrir la création artificielle de champs de gravitation par l'intermédiaire des ondes électromagnétiques, il y a lieu d'envisager des expériences susceptibles de permettre la détermination de la valeur de cette densité d'énergie diffuse ρ .

Dans ce but, il paraît opportun d'étudier l'action combinée d'un champ électrique E et d'un champ magnétique H , de valeurs élevées, perpendiculaires entre eux et variables en fonction du temps, sur une masse d'épreuve de faibles dimensions, constituée d'un matériau diélectrique qui n'entraîne que de très faibles variations dans les lignes de champ aussi bien électrique que magnétique. (fig. 15). En faisant passer rapidement le champ de la valeur initiale E_0 à l'instant zéro, à une valeur nulle atteinte à l'instant θ , nous pouvons écrire l'égalité :

$$\int_0^\theta (m + M) \gamma \cdot dt = \int_0^\theta \frac{m}{\rho} \cdot \frac{\partial}{\partial t} (E \wedge H) \cdot dt,$$

égalité dans laquelle M représente la masse de l'armature support de m .

En remarquant que γ est égal à $\frac{dv}{dt}$, nous pouvons procéder

lération de gravitation ayant pour expression; $\gamma = \frac{1}{\rho} \frac{\partial}{\partial t} (\mathbf{E} \wedge \mathbf{H})$.

Nous avons dit, en effet, (§ 6, p. 62) que l'induction, $\mathbf{B} = \mu_r \mu_0 \mathbf{H}$, semble correspondre, en réalité, à un champ magnétique qui résulte de la superposition du champ d'excitation $\mathbf{H}$ et des champs dus aux spins des atomes orientés sous l'action de $\mathbf{H}$.

Dans un barreau de ferrite, l'accélération de gravitation doit donc pouvoir prendre la valeur $\frac{1}{\rho} \frac{\partial}{\partial t} (\mathbf{E} \wedge \mathbf{B}) = \frac{\mu_r}{\rho} \frac{\partial}{\partial t} (\mathbf{E} \wedge \mathbf{H})$,

où μ_r représente la perméabilité relative du ferrite par rapport au vide de matière. Il peut être alors procédé à une expérience utilisant un équipage suspendu à un fil de torsion, par exemple, (fig. 16) dont le bras horizontal supporte, à ses extrémités, deux barreaux de ferrite polarisés en sens inverse. Ces barreaux peuvent, dans leurs déplacements, passer entre les armatures de deux condensateurs.

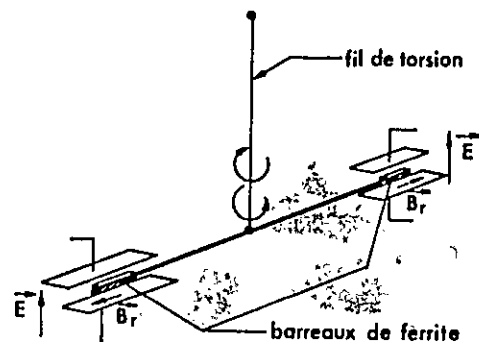


Fig. 16. — A chaque passage entre les plateaux des condensateurs, l'inversion du champ électrique fournit aux barreaux de ferrite, A et B, des impulsions de gravitation qui doivent produire des oscillations du pendule de torsion constitué par l'équipage suspendu.

En inversant le champ électrique à chaque passage des barreaux de ferrite entre les armatures des condensateurs, il doit être possible, par l'intermédiaire des impulsions de gravitation qui en résultent, de communiquer des oscillations entretenues à l'équipage suspendu par le fil de torsion.

Il importe de préciser que les expériences qui viennent, dans

leur généralité, d'être très sommairement décrites, n'ont pas, pour l'instant, fait l'objet de réalisations pratiques. Il serait surprenant, cependant, que le résultat n'en fût pas positif.

S'il existait des laboratoires où de telles expériences auraient été faites et gardées secrètes, disons à leur rencontre qu'il ne peut y avoir de secret pour une découverte aussi simple et objective et qu'elle se devait d'être largement et rapidement diffusée. La connaissance doit être mise à la disposition du plus grand nombre, et si elle ne l'était pas, elle ne pourrait avoir aucune valeur.

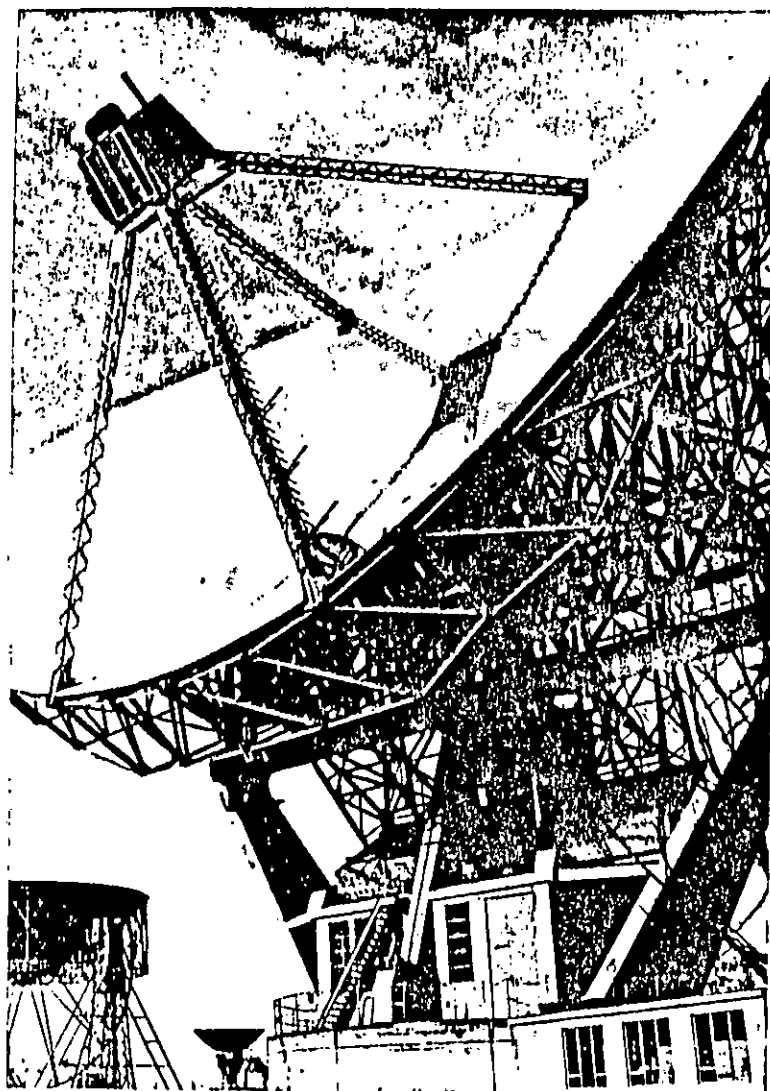
Nous avons vu qu'une onde électromagnétique transverse est toujours accompagnée d'un champ de gravitation dont les effets sont très faibles et probablement très difficilement mesurables. Ces effets sont masqués, d'une part, par une densité d'énergie diffuse ρ , sans doute considérable, et par l'importance prépondérante des actions électromagnétiques associées, d'autre part. Il est donc difficile de mettre en œuvre un mode de propulsion électrogravitationnel qui utiliserait des ondes électromagnétiques, car il faudrait faire appel, avec un rendement médiocre, à des énergies prohibitives.

Le calcul montre en effet, dans tous les cas, que la poussée obtenue, $P = \frac{1}{c} \frac{\partial W}{\partial t}$, est proportionnelle à l'énergie éjectée par unité de temps et inversement proportionnelle à la vitesse d'éjection. Cette dernière étant égale à la vitesse de propagation des ondes électromagnétiques, la poussée est réduite à son minimum pour une même énergie utilisée.

Afin d'obtenir une poussée de un kilo, par exemple, il faudrait éjecter, sous forme d'ondes électromagnétiques, une énergie de 300 Mégajoules par seconde; soit une puissance de 300 Mégawatts.

Nous pouvons en déduire que tout espoir de mettre en œuvre un moyen de propulsion économique, utilisant la Gravitation, suppose, au préalable, la découverte d'une possibilité d'action directe sur la répartition de la densité d'énergie diffuse ρ , sans qu'il y ait émission d'ondes électromagnétiques.

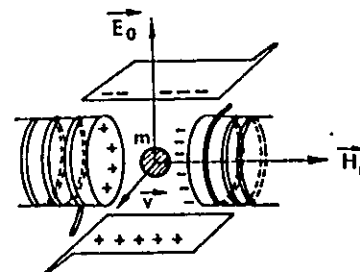
La découverte d'une telle possibilité ne peut, à notre avis, que résulter d'une connaissance plus approfondie des propriétés de la matière, compte tenu des éléments nouveaux que révèle la théorie développée qui repose sur l'existence de milieux énergétiques diffus.



PL. VII. — CET IMPOSANT RADIOTÉLESCOPE, INSTALLÉ A JODRELL BANK, NE POSSÈDE QU'UN ÉTROIT CANAL OUVERT DANS LE SPECTRE DE L'ÉNERGIE ÉLECTROMAGNÉTIQUE DIFFUSE QUI PARCOURT L'IMMENSITÉ DE L'UNIVERS.

Pour des longueurs d'ondes correspondant à des fractions d'angström, l'énergie diffuse non matérialisée semble ne pouvoir être détectée que par ses effets gravitationnels.

FIG. 15. — Les pièces polaires d'un électroaimant et les armatures chargées d'un condensateur créent un champ magnétique H_0 et un champ électrique E_0 . La décharge du condensateur fait disparaître le champ E_0 et doit communiquer à la masse m la quantité de mouvement :



$$(m + M)v = -\frac{m}{\rho}(E_0 \wedge H_0).$$

à l'intégration des deux membres de l'égalité précédente et obtenir entre les quantités de mouvement, la relation suivante :

$$(m + M)v = -\frac{m}{\rho}(E_0 \wedge H_0).$$

E_0 et H_0 étant perpendiculaires entre eux, nous pouvons, connaissant la vitesse v , les masses et les champs, déterminer la densité ρ cherchée :

$$\rho = \frac{m}{(m + M)} \cdot \frac{|E_0| \cdot |H_0|}{|v|}$$

La difficulté d'une telle expérience réside, non seulement dans la mesure de faibles vitesses, mais aussi dans le fait physique qu'il est difficile, à la limite, de distinguer une force de gravitation d'une action électromagnétique, puisqu'il s'agit, en réalité, des deux aspects d'un même phénomène.

Il faut éviter, en particulier, des résonances avec les pseudo-périodes de variation du champ qui peuvent se traduire par une dissipation calorifique dans la matière même de la masse d'épreuve et considérer, surtout, que l'expression, $\gamma = \frac{1}{\rho} \frac{\partial}{\partial t}(E \wedge H)$, n'est valable, en principe, que pour une onde électromagnétique transverse.

Il n'est pas impossible que l'utilisation de barreaux magnétiques de ferrite permette plus facilement de mettre en évidence l'accé-

LES PERSPECTIVES D'UNE THÉORIE ÉNERGÉTIQUE QUANTIQUE ET GRAVITATIONNELLE

Il ne pouvait s'agir, dans cet ouvrage, d'étudier en détails tous les phénomènes qui se trouvent expliqués par l'existence des milieux énergétiques et la loi de matérialisation.

Le fait de retrouver, en partant des lois de l'électromagnétisme classique, les équations de la Relativité, celles de la Mécanique ondulatoire et de la Gravitation, sont des éléments que nous avons jugés suffisants pour affirmer que toutes les vérifications expérimentales relatives à chacune de ces théories pouvaient, globalement, être portées à l'actif d'une théorie générale qui les confirme.

Il existe, cependant, des résultats particuliers et nouveaux à retenir, qui sont, en résumé, les suivants :

— La constante de Planck, et la vitesse de la lumière qui est égale à la racine carrée du potentiel de Gravitation, ne sont pas des constantes universelles; alors que la perméabilité μ_0 du vide de matière peut être considérée comme telle,

$$h = 8\pi k_0 \mu_0 q^2 c$$

et

$$c = \sqrt{V_g} = \frac{1}{\sqrt{\epsilon \cdot \mu_0}}$$

— L'énergie équivalente à la masse, $W = mc^2$, dont l'expression fut établie par Einstein, et démontrée dans sa généralité (§ 2, p. 15) sans faire appel aux principes de la Relativité, se confond

avec l'énergie potentielle de Gravitation de la masse m , $W = mV_g$, en une seule et même expression,

$$mV_g = mc^2 = \frac{m}{\epsilon \mu_0}$$

— toute onde électromagnétique transverse est accompagnée d'un champ de gravitation qui a pour expression,

$$\gamma = \frac{1}{\rho} \frac{\partial}{\partial t} (E \wedge H)$$

— pour de faibles variations, le potentiel de gravitation, V_g , obéit à l'équation de propagation :

$$\Delta V_g - \frac{1}{V_g} \left(\frac{\partial^2 V_g}{\partial t^2} + 4\pi k \rho_m \right) = 0$$

Notons que la sommation vectorielle de champs de gravitation, ou celle de champs électriques ou magnétiques, n'est possible, en première approximation, que dans le cas de faibles variations de la permittivité ϵ , parce que ces champs participent à l'existence même du milieu énergétique de propagation.

La découverte la plus importante est certainement celle de milieux constitués d'énergie électromagnétique diffuse à haute concentration, responsables des forces d'inertie et des accélérations gravifiques; découverte qui ouvre la voie à de nouvelles recherches.

L'existence de ces milieux diffus est confirmée par les champs électromagnétiques et les champs de gravitation qui sont des éléments mesurables et qui traduisent, avec certitude, l'existence d'une énergie utilisable.

Les masses d'eau qui dévalent d'un barrage de retenue, par le canal des conduites forcées, vers l'usine hydroélectrique, utilisent l'énergie diffuse de gravitation pour actionner les turbines. Cette énergie diffuse est alors transformée, canalisée puis transportée, à très basse fréquence, autour de conducteurs « haute tension » qui utilisent la propriété des particules chargées de pouvoir conden-

ser ou véhiculer cette énergie avec de faibles pertes. Il est connu, en effet, que l'énergie électrique transportée par un conducteur se trouve, en majeure partie, localisée, sous forme diffuse, dans l'espace extérieur au conducteur et que l'utilisation de champs électriques élevés, correspondant à de hautes tensions, permet d'augmenter la proportion de cette énergie extérieure avec, pour conséquence, une diminution des pertes dans le conducteur sous forme thermique (Effet Joule).

L'existence de cette énergie non-matérialisée, et qui n'est pas thermodynamique, est également confirmée par la dématérialisation des neutrinos et par le rayonnement cosmique diffus qui prend naissance au sein des espaces intersidéraux où, comme nous l'avons vu, la densité d'énergie diffuse est la plus élevée.

Essayons d'imaginer comment peut naître une particule cosmique.

Quand le champ électrique tend, en un point de l'espace et au hasard de ses fluctuations, à dépasser la valeur limite ϵ_d , il doit se produire une véritable « implosion » d'énergie. Les zones disruptives qui apparaissent, doivent provoquer une augmentation de la permittivité ϵ et, par suite, une déformation des lignes de force du champ électrique. La concentration qui en résulte entraîne une dépression d'énergie diffuse au voisinage du photon cosmique ainsi créé, lequel est projeté à la vitesse $\frac{1}{\sqrt{\epsilon\mu_0}}$ dans la direction

de l'onde diffuse qui lui a donné naissance, tout en prélevant une énergie proportionnelle à sa fréquence moyenne, $W = h\nu_m$, au chaos diffus environnant. La matérialisation cosmique se présenterait donc comme un véritable phénomène de « Négentropie ».

Si la répartition de l'énergie diffuse est aléatoire, il doit lui correspondre une courbe de distribution dont le relevé, déterminé expérimentalement, permettrait de connaître la densité d'énergie la plus probable dans chaque domaine de fréquence ν , $\nu + d\nu$. Cette courbe de distribution constitue, peut-être, la clef du mystère de la stabilité des particules matérielles élémentaires, expliquant la Radioactivité naturelle autant qu'artificielle.

Pourquoi les particules stables, électrons, protons noyaux atomiques, et leurs anti-particules, ne correspondraient-ils pas à des pics énergétiques de cette courbe particulière ? (fig. 17).

La stabilité n'est peut-être, en somme, que la traduction d'un phénomène de résonance qui résulte de l'interaction d'une

particule avec le milieu diffus pour une fréquence correspondant à l'un des pics de la courbe de distribution.

Et pourquoi les pics ne seraient-ils pas distribués selon la répartition d'une décomposition en série de Fourier ?... avec deux fréquences moyennes fondamentales se modulant l'une l'autre : celle du proton ν_p et celle du neutron ν_n . Le terme général de cette intermodulation aurait très approximativement pour expression :

$$A_{k_0} \cos(k_0 \nu_p t - \varphi_0) \cdot A_{k_1} \cos(k_1 \nu_n t - \varphi_1)$$

soit :

$$1/2 A_{k_0} \cdot A_{k_1} \{ \cos[(k_0 \nu_p + k_1 \nu_n)t - (\varphi_0 + \varphi_1)] + \cos[(k_0 \nu_p - k_1 \nu_n)t - (\varphi_0 - \varphi_1)] \}.$$

Si le produit $A_{k_0} \cdot A_{k_1}$ est maximal pour $k_0 = k_1$, cela peut expliquer la stabilité des noyaux dans lesquels le nombre de nucléons se répartit également entre neutrons et protons. Cela pourrait expliquer également l'existence et la stabilité des électrons et des deutérons, correspondant à $k_0 = k_1 = 1$.

Les décalages de pics dus aux superpositions pourraient expliquer aussi l'existence des énergies de liaison. La probabilité de désintégration d'une particule d'énergie, $E = h\nu$, serait alors d'autant plus faible que la dérivée correspondante $\frac{dE}{d\nu}$, sur la courbe de distribution d'énergie diffuse, serait plus faible.

Cela devrait entraîner l'existence, entre les pics, de *particules métastables* dont les durées de vie pourraient accuser d'assez grandes variations suivant les fluctuations du milieu. Ces particules ne mériteraient-elles pas alors, d'être appelées « particules étranges » ?... Pourquoi ne correspondraient-elles pas aux « mésons » dans la portion de courbe située entre le pic de l'électron et celui du proton, et aux « baryons » entre le pic du proton et celui du deutéron ?

Si cela était, les Physiciens nucléaires devraient découvrir, à l'aide des accélérateurs de très haute énergie actuellement en construction, de nouvelles particules métastables, des sortes « d'hyper-baryons », se situant entre le pic du deutéron et celui de la particule α , et même au-delà. Le « triton » étant, parmi ces particules, celle qui possède, probablement, la plus grande durée de vie.

Il n'est pas impossible que la radioactivité naturelle soit, finalement, la conséquence d'une instabilité due à la faible amplitude

relative du pic énergétique qui correspond à un noyau radioactif, comparée à celle des fluctuations statistiques du milieu ambiant.

Les hypothèses qui viennent d'être faites ne sont que des suggestions susceptibles d'orienter les recherches vers une interprétation satisfaisante des phénomènes nucléaires. Nous pensons, cependant, que l'étude de ces phénomènes en rapport avec la structure intime de la matière et la physique en général, comme avec l'Astronomie et la Cosmologie, ne peuvent plus, désormais, ignorer l'existence des milieux électromagnétiques diffus qui semblent constituer la trame de notre Univers, nous obligeant, dans certains domaines, à une révision profonde de nos conceptions scientifiques.

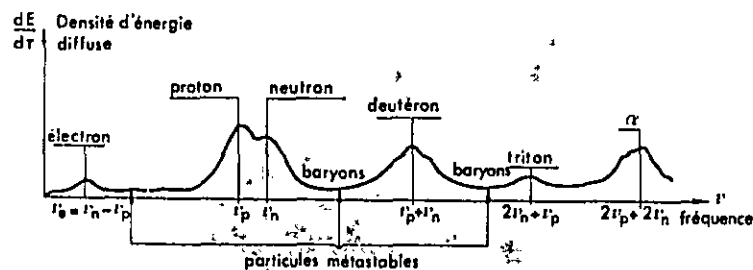


FIG. 17. — Nous pouvons imaginer une courbe de distribution donnant la densité la plus probable d'énergie électromagnétique diffuse en fonction de la fréquence, et dans laquelle, les pics correspondraient aux particules et noyaux atomiques les plus stables. Dans cette courbe le pic de l'électron aurait pour abscisse $\nu_e = \nu_n - \nu_p - d\nu_0$. La différence $h d\nu_0$ correspond à l'énergie que l'on associe au neutrino et qui traduit simplement l'échange d'énergie avec le milieu diffus.

Nous vivons certainement dans un monde en équilibre précaire dont il nous faut, pas à pas, découvrir tous les liens secrets qui en constituent la réalité causale, si nous voulons avoir une chance d'assurer l'avenir et la pérennité de notre espèce menacée par des limites, sans doute très étroites, dans les conditions naturelles de survie.

Si, par exemple, la fonction de distribution de l'énergie diffuse est, comme il semblerait, responsable de la stabilité de la matière, il est sûr que le moindre décalage de fréquence relativement aux pics de cette fonction, peut entraîner un effroyable cataclysme qui transformerait, presque instantanément, notre système solaire en une gigantesque nova, ou même une supernova, où plus d'un



PL. VIII. — AMAS DE GALAXIES VUES AU GRAND TÉLESCOPE DU MONT PALOMAR DANS UNE INFIME PARTIE DE L'HÉMISPHERE BORÉAL. Ces multitudes d'Univers-îles, qui semblent peupler l'immensité du Cosmos, correspondent à des condensations locales fantastiques d'énergie matérialisée, bien que la densité d'énergie diffuse y soit encore très élevée dans les régions interstellaires.

122 ÉNERGÉTIQUE QUANTIQUE ET GRAVITATIONNELLE

million d'années d'aventure humaine viendrait s'abîmer, sans qu'aucune trace n'en subsistât, dans un fantastique embrasement. Cela s'est déjà produit et se produit encore dans toutes les galaxies, bien que la probabilité d'un tel bouleversement soit très faible.

Pour éviter d'être les victimes ignorantes d'un sort aveugle et implacable, pour avoir quelque chance d'agir sur notre destin, nous devons prévoir — et prévoir, c'est d'abord connaître. Et cela nécessite un ensemble de connaissances à la base, pragmatiques, simples et cohérentes.

Rappelons que la théorie d'énergétique quantique et gravitationnelle repose, essentiellement, sur la possibilité d'une représentation électromagnétique de toute densité de quantité de mouvement,

$\frac{\partial mv}{\partial \tau} = \mathbf{D} \wedge \mathbf{B}$, représentation suggérée, à l'origine, par l'existence du vecteur de Poynting. Ce lien, entre la mécanique classique et l'électromagnétisme, nous fait découvrir alors que l'ensemble des concepts de la mécanique, de l'optique et de la physique en général, matière, énergie, potentiels, photons ou particules, peuvent être exprimés en termes électromagnétiques établis à partir des équations fondamentales de Maxwell-Ampère et de Maxwell-Faraday.

Ne s'agit-il pas, en fait, d'une véritable théorie unitaire ? qui place l'électromagnétisme et l'électronique à la base de l'existence même de la matière et de notre Univers sensible et observable.

Il serait, cependant, peu raisonnable de dire que cet Univers n'est doué que de propriétés électromagnétiques, parce qu'il ne nous est pas possible d'exprimer scientifiquement la nature des choses. Mais nous pouvons affirmer par contre, parce qu'il s'agit de rapports, que toutes les lois physiques actuellement connues doivent pouvoir s'exprimer en termes qui relèvent de l'électromagnétisme classique.

Notons qu'il n'existe, quant aux résultats expérimentaux, aucun conflit entre la théorie proposée, la Mécanique classique, la thermodynamique, la théorie quantique, la Mécanique ondulatoire, la Relativité restreinte ou généralisée, que nous pouvons même, à tout instant, invoquer comme preuves des hypothèses qui ont été avancées. Nous devons, pourtant, rejeter de ces théories certains principes a priori. Non pas que nous pensions devoir les juger faux ou discutables, mais parce qu'ils sont devenus

simplement inutiles et parfois nuisibles à la compréhension de l'ensemble.

Nous avons montré, à l'aide des équations de Maxwell et des transformations de Lorentz qui en sont issues, sans faire appel aux principes relativistes, qu'à un milieu énergétique, en général, pouvait être associée une masse d'inertie que nous avons calculée (§ 2, p. 15). Nous avons démontré, en utilisant la loi de matérialisation, l'équation fondamentale de la Mécanique quantique (§ 4, p. 29) et celle de la Mécanique ondulatoire (§ 5, p. 46) qui légitiment l'équation de Schrödinger dont les limites de validité ont été précisées. Nous avons établi, finalement, les équations des champs de gravitation, sans faire appel à la Relativité Générale et en rejetant même le principe d'une vitesse de la lumière, « Constante universelle ».

Nous ne doutons pas un seul instant, car c'est une réalité indéniable, que la Relativité pourrait parvenir aux mêmes résultats, à condition, toutefois, d'y introduire les lois de l'électromagnétisme qu'il n'est pas possible de considérer comme une géométrisation de l'espace. Mais nous ne voyons pas l'intérêt qu'il y aurait à maintenir des principes, dont l'utilité ne se justifie plus, qui n'entraînent que des complications, se traduisent par des conceptions peu familières et risquent de conduire à des erreurs engendrées par le brassage d'idées inhabituelles, abstraites et parfois déroutantes, où notre logique finit par se perdre dans les dédales de l'incertitude et des paradoxes.